GREEN CLIMATE FUND

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GCF Water Project Design Guidelines

Part 1: Practical guidelines for designing water-climate resilient projects

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Abbreviations

ADB	Asian Development Bank
AE	Accredited Entity
AIS	Administrative and Institutional System
CC	Climate Change
CMIP	Coupled Model Intercomparison Project
CR	Climate Resilient
CRIDA	Climate Risk Informed Decision Making
DAE	Direct Access Entity
DAPP	Dynamic Adaptive Policy Pathways
ESIA	Environmental and Social Impact Assessment
GCF	Green Climate Fund
GCM	Global Circulation Model
GWP	Global Water Partnership
IAE	International Access Entity
IPCC	Intergovernmental Panel on Climate Change
IWRM	Integrated Water Resource Management
JICA	Japan International Cooperation Agency
MCEM	Multi Criteria Evaluation Methods
NAP	National Adaptation Plan
NDA	National Designated Authority
NDC	National Determined Contribution
NRS	Natural Resource System
OECD	Organization for Economic Cooperation and Development
RCM	Regional Climate Model
RCP	Representative Concentration Pathways
SDG	Sustainable Development Goal
SES	Socio-Economic System
TA	Technical Assistance
WASH	Water Sanitation and Hygiene
WMO	World Meteorological Organization
WRS	Water Resource System (= NRS-SES-AIS)
WSSG	Water Security Sectoral Guide

Glossary and Terminology

Term	Description
Adaptation	*Adaptation* is any adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects which moderates harm or exploits beneficial opportunities.
Adaptation pathway Alternative term use: adaptation strategy	*Adaptation pathway* is a sequence of measures in time to achieve a set pre-defined objectives under changing external changing conditions, such as climate, socio-economic factors or other developments.
IWRM	IWRM is a cross-sectoral approach at the core of UN-SDG6, designed to promote the coordinated development and management of water, land, and related resources to maximise economic and social welfare equitably, without compromising the sustainability of ecosystems and the environment
Resilience	*Resilience* is the capacity for a socio-ecological system to: (1) absorb stresses and maintain function in the face of external stresses imposed upon it (e.g. by climate change), and (2) adapt, reorganize, and evolve into more desirable configurations that improve the sustainability of the system, leaving it better prepared for future (climate change) stresses.
Indicators (evaluation)	*Indicators* express to what extent the objective has been achieved. As such, they are the measuring rod that show how successful a project (or project portfolio) has been.
Flexible (measures) Alternative term use: adaptive	*Flexible measures* can be adapted (e.g. intensification of the action), abandoned (switch to a different action) or extended (add an action) at relatively low cost or having small societal impact. Flexible actions do not result in *lock-ins* and have little influence on potential future options (i.e. have less *path-dependencies*).
Lock-in	*Lock-in*: Situation where some future measure in a pathway can only be implemented with associated high costs or high societal impact.
Measure Alternative terms used: intervention, action	A *measure* is a particular course of action. Types of measures are: technical (structural), ecological, economic, regulatory, institutional.
Mitigation	*Mitigation* is reducing the flow of heat-trapping greenhouse gases into the atmosphere, either by reducing sources of these gases or enhancing the "sinks" that accumulate and store these gases.
Objective Alternatives used: goals, aim	An *objective* defines what is to be achieved or how some target is to be met. Goals identify needs, prioritize issues and define targets and constraints on the actions to be taken to meet the targets. Goals may define preferred courses of action. For example, the goal might be to apply user-oriented demand management measures rather than relying on large-scale water supply infrastructure development.
Path dependency	*Path-dependency* is the reduced flexibility between actions, due to the commitment to certain development trajectories and/or limited future alternatives to adapt to unforeseen changes. Path dependency may result from the legacy of the infrastructure and/or the associated economic and social developments triggered by the investment (e.g., higher costs or limits).

	Decision path-dependency arises from sequencing decisions and *scenario path-dependency* arises from environmental or socio-economic changes in the operating environment (e.g., population growth reduces space to adapt).
Robustness (of a system)	The ability of a system to remain functioning under a large range of disturbance magnitudes (i.e., plausible scenario conditions).
Robust measures	*Robust measures and/or strategy:* Measures that result in acceptable indicator values under a wide variety of futures.
Scenario	A *scenario* is defined as the environment exogenous to the water system under consideration that cannot be controlled by the decision maker involved. Socio-economic developments and climate change are the most important scenario components in water resources systems.
Strategy Alternatives used: plans, portfolio plan, portfolio of projects, policy, adaptation pathways Sometimes the term Scenario is used; this should be avoided as scenario's describe external conditions	A *strategy* is defined as a logical combination of individual measures or decisions that accomplishes the stated objectives and satisfies the constraints imposed on the water resources system. A strategy can be described in terms of (alternative) pathways.
Water Resource System	The *Water Resource System (WRS)* is the integrated system of the Natural Resource System (NRS), Socio-economic System (SES) and the Administrative and Institutional System (AIS).
Water security	The vision for a paradigm shift in water security is to secure water resilience and water services under conditions of increased climate change impacts. Water security is part of the health, food, and water security result area emphasises mostly adaptation projects to harness available water resources and develop resilient water infrastructures and services.

Note: For an overview of general key terms as used by GCF, reference is made to the Glossary of the GCF Programming Manual.

Executive summary

Project proposals, submitted to GCF for funding, should be based on a climate rational, sound technical and economic analysis, innovation, as well must comply with the specific conditions of GCF. These guidelines provide practical information and recommendations on how to prepare the technical content of water related project proposals. These guidelines are complementary to the Water Security Sectoral Guide[1] that describes the position and ambitions of GCF in the water sector as well as the financial mechanisms and implementation arrangements that GCF is willing to support. This document translates the Water Security Sectoral Guide into how projects can be designed to be suitable for GCF to finance, because they meet GCF's investment criteria. The specific GCF requirements and funding proposal template are described in the GCF Programming Manual.

The **target audience** of these guidelines are the Direct Access Entities and Accredited Entities. The guidelines might also be useful for International Access Entities, or any other organisation interested in accessing GCF funding.

The technical analysis as described in these guidelines starts with recognizing the intricate connections between the natural resource system, the socio-economic system (the users of the water) and the administrative and institutional system. These connections require that water sector projects should follow an **integrated approach** to reduce climate risk. How much 'integration' is needed depends on the specific project, but each proposed water sector project should consider how the project might impact other users, sectors and the environment. Moreover, these Guidelines recommend a strong **engagement of all relevant stakeholders**. To ensure such integrated approach and the involvement of all stakeholders, a system based **structured process of project design** needs to be followed.

GCF funding requires that project proposals have a clear **climate rationale,** in line with policy paper B.33/05**,** to ensure that the project responds directly to climate change challenges. The climate rationale is embedded in the structured process of project design. This will provide the evidence of the impacts of climate change by identifying the hazards, exposure and vulnerability of the system under consideration. This **risk analysis** provides the information based upon which a set of alternative interventions can be formulated, and the preferred solution (the project/program) is selected and proposed to GCF for funding.

GCF has adopted a four-pronged approach to accelerate and scale up climate innovation. The first prong focuses in establishing a conducive policy and institutional environment for climate solutions. The second is to catalyse and facilitate the emergence of climate innovation. The third is to use scarce public resources and de-risk investments that have potential to foster private finance. The fourth is to accelerate the widespread adoption of new climate solutions.

Water is a cross-cutting with many other GCF climate sectors such as agriculture, food, forests, energy (hydropower), urban, ecosystems management and infrastructure. This document focusses on providing project design guidelines by establishing a structured process and defining the main steps for project development (Part 1) and providing applications for IWRM and three water sub-sectors (Part 2), climate resilient WASH, drought risk management and flood risk management.

IWRM projects. Only IWRM projects (mainly focussed on adaptation) that include actual investments can be considered for regular GCF funding. IWRM planning projects (such as river basin management and plans) and feasibility studies can be supported by GCF by means of GCF's Readiness and Preparatory Support Program. IWRM projects eligible for GCF funding should include components that bring together water users from multiple sectors to co-invest in mutual benefits. This can include users from other sectors of GCF such as agriculture, food, forests, energy, urban, ecosystems management and infrastructure.

Climate Resilient WASH projects. WASH projects should aim at improving the health, socio-economic situation and resilience of the communities involved preferably by addressing the whole WASH system (cross-cutting project between adaptation and mitigation). The projects should ensure that the proposed interventions make

[1] https://www.greenclimate.fund/document/sectoral-guide-water-security

the WASH infrastructure and services sustainable, safe and resilient to climate-related risks. These resilient WASH systems should contribute to helping build community resilience to the impacts of climate change and by improving water and energy efficiency. In summary, these projects should deliver both mitigation (e.g., hydropower or use of renewable energy for water production) and adaptation benefits.

Drought Risk Management projects. Drought risk projects should aim to reduce the risk of communities to water shortages and to prevent severe economic, social and ecological impacts of droughts. GCF drought projects must address the exacerbation of drought due to climate change. Drought projects should include an extensive analysis of the historical hydro-meteorological condition (e.g., water availability, depth-duration, precipitation) and the expected changes due to climate change. Hydrological/hydraulic (1D-2D) and/or water balance modelling might be required to assess the impact of climate on the users and to evaluate the effectiveness of interventions that aim to reduce the risk. Interventions can address the drought and other cascading risk and consequences. Nature Based Solutions (NBS) should be considered, where possible, as they are often cheaper and more sustainable. Also, consideration should be given to cross sectoral interventions, for instance water allocation across sectors including priority settings, during drought events.

Flood Risk Management projects. Flood projects should aim at reducing the impacts of flooding. GCF flood projects should address the increased risk of flooding (and related damage) due to climate change. As with drought projects this will require an extensive analysis of the historical hydrology and the expected impacts of climate change on the flooding situation. Hydrological and hydro-dynamic models are needed to determine the potential flooding frequency and extent. These models, together with impact models (damages, benefits) are also needed to determine the effectiveness of interventions that aim to reduce the risk. Possible interventions should consider the full risk cascade and include the reduction of the hazard, increased protection, regulation of land-use, and the raising of awareness and preparedness. As with droughts, Nature Based Solutions are promising interventions in flood projects, both for hazard reduction as well as for protection against floods.

1 Introduction

1.1 The GCF Water Security Sectoral Guide as a starting point

This document provides practical guidelines for developing projects and programmes that build climate resilience by enhancing water security, in line with GCF's strategy and mandate. **GCF's goal statement for water security is:** "GCF promotes a paradigm shift in water security that is low-carbon, resilient to climate change, and meets the goals of the UNFCCC and Paris Agreement". The water system is the connector to other sectors. Water security is achieved with climate, energy, food and ecosystem securities (see Figure 1).

Figure 1: Water security as connector to other sectors



Source: Amgad Elmahdi, 2022[2]

These guidelines are complementary to the Water Security Sectoral Guide[3] that describes the position and ambitions of GCF's investment in the water sector as well as the financial mechanisms and implementation arrangements that GCF is willing to support.

Figure 2: Possible actions for each paradigm shifting pathway following the four pillars of the GCF Strategic Plan

[2] https://unece.org/environment/documents/2022/10/presentations/role-gcf-water-security-transboundary-lens-mr-amgad

[3] https://www.greenclimate.fund/document/sectoral-guide-water-security

Sector		Actions across the drivers of the GCF Strategic Plan			
Water security		Transformational planning & programming	Catalyzing climate Innovation	Mobilization of finance at scale	Coalitions & knowledge to scale up success
Paradigm shifting pathway	Enhancing water conservation, water efficiency, and water re-use	• Encourage benchmarking across service sectors and providers • Incentivise circular economy approaches for resources recovery • Design low emission climate resilient investment pathways that maximise long term water benefits • Strengthen water security into NDCs and NAPs	• Mainstream climate smart water and agriculture using digital solutions • Promoting new asset classes in sanitation and water re-use that follow these characteristic in finance, revenue stream, SDGs and Paris agreement • Reduce performance uncertainty through asset management • Advocate decenetralised water supply and wastewater management • Support desalination using renewable energy • Employ data science and initiate "big data" solutions, such as reducing CO2 emissions	• Allocate grant funding for technical assistance and capacity building • Enable private sector participation by supporting credit enhancements and full cost recovery through direct and indirect charges • Support comprehensive cost-benefit analyses with co-benefit from EbM • Introduce tax initiatives to contain adverse environmental impact of activities • Support carbon credits intiatives for revenue generation	• Improve available information and data acquisition through knowledge platforms • Support peer-to-peer learning and regional exchanges of lessons learned and best practices • Support catchment-based initiatives to promote water stewardship • Empower communities into the decision-making process • Foster cultural-specific communication with stakeholders • Encourage behaviour change in water conservation practices
	Strengthen integrated water resources management – protection from water-related disasters, preserve water resources and enhanced resilient water supply and sanitation services	• Link IWRM with water safety plans for long term adaptive planning to map climate hazards with risks and vulnerability • Encourage transboundary water resource cooperative arrangements • Support resilient planning and design processes dealing with uncertainties • Improve flood modelling into disaster risk resilience assessment and drought rationale through climate vulnerability assessment • Integrate social and gender sensitive dimensions into water security interventions	• Promote and implementing a well-managed mix and integration of the Grey-Green Infrastructure to enhance the adaptability and resilience of coastal and upstream communities • Mainstream rainwater catchment harvesting and storage systems • Promote stromwater harvesting (e.g., sponge cities, agriculture) • Advocate for water re-use and water recycling as alternative water sources • Strengthen resilient WASH programmes preventing maladaptation • Introduce EbM to enhance climate resilience in water infrastructure and build coastal resilience • Contribute to urban climate resilience for flood and land use management • Reduce drought vulnerabilities through water re-use and recycling	• Catalyse public funds to scale-up blended finance • Enhance projects' risk-return profile • Address risks vs. perceived risks • Improve creditworthiness through credit enhancements and de-risking • Defer investment using resilient water solutions within the whole water cycle • Expand micro-finance to support household level resilient water systems • Initiate ecosystem-based insurance and disaster risk insurance and bonds • Participate in specialised water and blue-green funds	• Enhance collaboration with independent institutions for Monitoring, Evaluation and Verification • Promote partnerships and new alliances in water security • Strengthen innovation climate hubs • Develop localised accredited climate education programmes • Invest in research to support evidence-based decision making

Source: GCF Water security Sectoral Guide

The Water Security Sectoral Guide outlines two major pathways for paradigm shifts:

- Pathway 1: Enhance water conservation, water efficiency and water reuse. E.g., through demand management, smart digital water management, decentralized operation models and resource recovery.
- Pathway 2: Strengthen integrated water resources management – protection from water-related disasters, preserve water resources and enhanced resilient water supply and sanitation services. e.g., through Ecosystem-based management, alternative water sources and IWRM.

It describes **what** is to be achieved. This document, the Water project design guidelines, translates the Water Security Sectoral Guide into **how** projects can be designed to be suitable for GCF finance, because they meet GCF's investment criteria and deliver results in GCF's result areas. It addresses in a structured way how to support GCF's **strategic pillars** for water security for example by:

- Enhancing transformational planning by including long term sustainability considerations into project design.
- Catalysing innovation by stimulating the analysis of a wide portfolio of solutions during project preparation
- Supporting the mobilization of finance through transparent project evaluations demonstrating clear societal benefits and return on investments
- Helping the upscaling of success by offering a comprehensive evidence-based method that can support capacity building and deliver best practice examples.

It also enforces many **enablers** to achieve the paradigm shifts that were identified in the Water Security Sectoral Guide. A structured process is provided on how to apply scientific evidence for different decision-making steps (Enabler 1). It offers a collaborative approach for project development according to Integrated

Water Resources Management (IWRM) principles that will target local context specific objectives and indicators to measure success (Enablers 3 and 4).

1.2 Scope of the water project design guidelines

The **target audience** of this Guideline document are the Direct Access Entities (DAE) at the national levels, who co-originate projects with the National Designated Authorities (NDA). It is assumed that the guidelines will also be helpful for the International Access Entities (IAE) and/or useful to any organisation interested in accessing GCF funding for water projects.

GCF water related activities are addressed at various levels. For these Guidelines we distinguish:

1. **National Adaptation Plan (NAP).** The NAP identifies medium- and long-term adaptation needs and developing and implementing strategies and programmes to address those needs. Water is often a prioritised sector in these NAPs.
2. **Country programme**. Country programmes set out the country's objectives for engaging with GCF and provide an indicative list of proposed projects (in various sectors, not only the water sector) that the country intends to submit for GCF funding. GCF encourages countries to develop and submit country programmes so that the linkages and priorities among projects of a particular country are clear. This is the responsibility of the NDA.
3. **Water programme**. A programme is defined here as a set of related projects and activities aimed at achieving one or more strategic objectives. An example is a wastewater recycling programme that provides on-lend funds to different municipalities in the same country, or a transboundary water management programme that funds interventions in different countries sharing a water resource.
4. **Individual water 'project'**. A project is a set of activities carried out in a temporary partnership, aimed at achieving predefined results within certain preconditions (e.g., time, money). These projects will often have specific requests for investments.

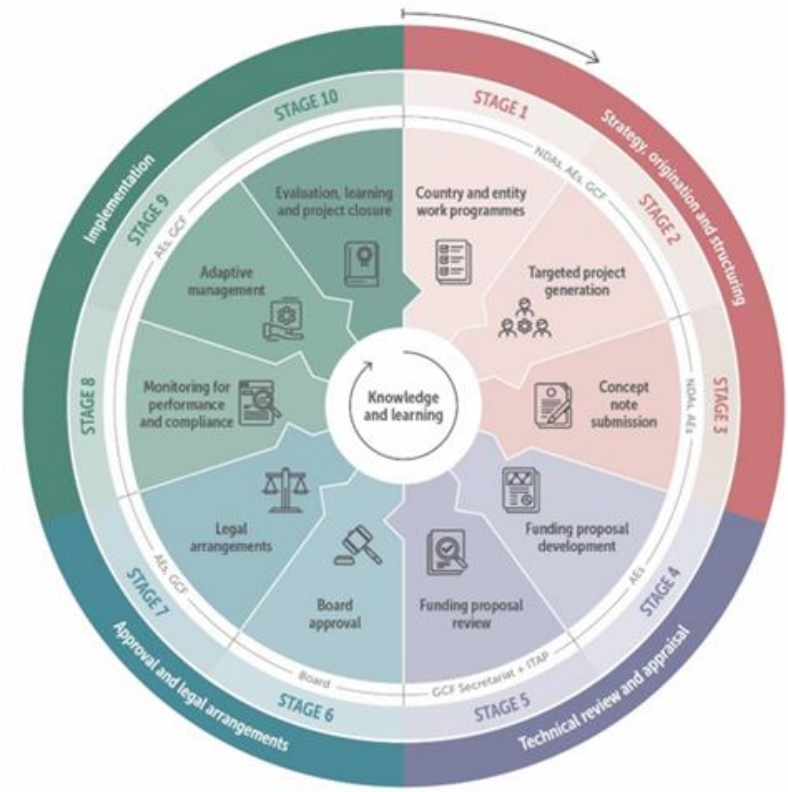
This document addresses mainly the last two categories, ensuring that project proposals submitted to GCF for funding are technically sound and meet GCF's investment framework. The first two categories might benefit from the Guidelines when identifying water-related projects to be included in their programmes and plans. Developing NAPs and Country Programs can be financed by GCF's Readiness Programme[4].

Water is only one of several sectors under the eight results areas of the GCF. Due to the integrated nature of the water sector, there are many linkages with other GCF climate sectors such as agriculture, forests, energy (hydropower), urban, ecosystems management and infrastructure.

This document aims to support better project design, in particular the conceptualization of effective solutions to address **climate change induced problems.** These guidelines will support the stages 3 (Concept note submission) and 4 (Funding proposal development) in the Project cycle of GCF, as illustrated in Figure 3.

Figure 3: GCF Project Cycle

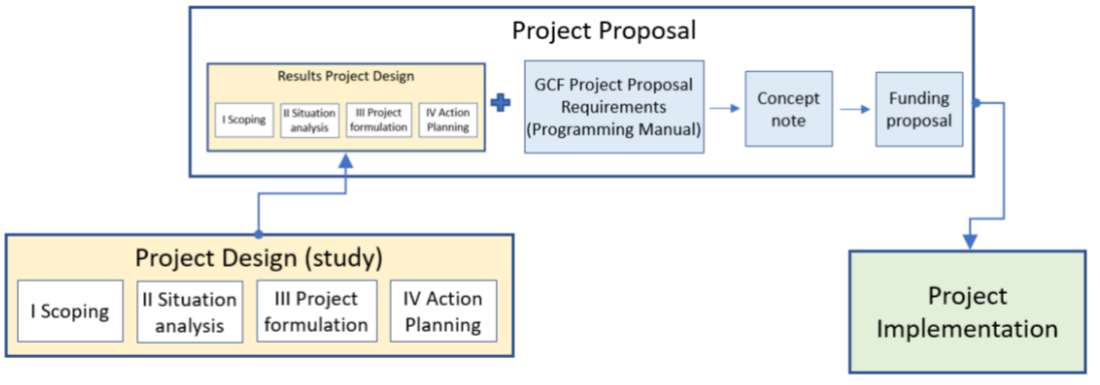
[4] https://www.greenclimate.fund/document/readiness-and-preparatory-support-guidebook



Source: GCF Programming Manual[5]

This document will describe how a project can be conceptualized by following a structured number of phases. These phases of project design will be described in chapter 2. After the project is properly conceptualised, the next step is to translate the project proposal in the GCF project proposal format as described in the GCF programming manual[6]. The project proposal will include the results of the project design as illustrated in Figure 4. Project proposals are delivered in two stages to the GCF, first a **concept note** which after approval can be further expanded into a full **funding proposal**.

Figure 4: Position of project design in developing a project proposal for GCF



[5] https://www.greenclimate.fund/document/programming-manual

[6] https://www.greenclimate.fund/sites/default/files/document/gcf-programming-manual.pdf

1.3 Linking project indicators to investment criteria

An additional consideration in designing a project is to give specific attention to GCF's investment framework[7]. The GCF's investment framework defines **six investment criteria** and related indicators that provide guidance in the development, assessment and approval of projects. They seek to promote consistency and transparency in funding proposals and promote efficiency in the assessment process. By enhancing the clarity of how funding proposals meet GCF investment criteria, these indicators can be used to enhance the quality of funding proposals. An overview of the GCF Investment criteria and how they can be advanced from a project design perspective is given below. The reader is directed to the specific GCF document[8] on this subject for a more detailed description.

- **Impact potential** - Potential of the project or programme to contribute to the achievement of GCF's objectives and result areas.
    - ➔ Impact potential relates mostly to the effectiveness of the proposed investments to reduce climate change impacts (e.g. expected change in loss of lives, values, livelihoods, etc) and lower GHG emissions.
- **Paradigm shift potential** - Degree to which GCF can achieve sustainable development impact beyond a one-off project or programme investment through replicability and scalability.
    - ➔ Paradigm Shift potential basically asks for contribution of the proposed investments to either pathway 1 or 2, though specific projects might identify other paradigm shifts.
- **Sustainable development potential** – Degree to which the actions align with national SDG priorities and deliver wider benefits and priorities.
    - ➔ For proper project design relevant co-benefits in economics, social, environment and gender empowerment should be part of the assessment.
- **Needs of the recipient** – Vulnerability and financing needs of the country and population.
    - ➔ For project design this means that a vulnerability analysis is needed to identify priority areas for interventions. A description of financial, economic, social and institutional needs and the barriers to access sources of climate-related finance is also required.
- **Country ownership** - Beneficiary country ownership of, and capacity to implement, a funded project or programme (policies, climate strategies and institutions).
    - ➔ The final project design should be aligned with NDCs, relevant national plans, indicators, and/or enabling policy and institutional frameworks. It should be explained how engagement of relevant stakeholders will increase ownership.
- **Efficiency and effectiveness** - Economic and, if appropriate, financial soundness of the programme/project: Does the project foster cost-effectiveness and private sector funding mobilisation?
    - ➔ These GCF criteria go beyond the more regularly used definitions of effectiveness and efficiency in project design. Indeed, the project needs to be effective in delivering impact (impact potential) against low net costs. But in addition, it is important to seek an optimal ratio of co-financing (concessional, private sector contribution), expected rate of return, and application of best practices.

———————————————

[7] https://www.greenclimate.fund/projects/investment-framework

[8] https://www.greenclimate.fund/sites/default/files/document/investment-criteria-indicators.pdf and https://www.greenclimate.fund/projects/criteria

Most of these GCF investment criteria and indicators coincide with the direct project operational objectives and indicators as described above. More suggestions on how to include links between the relevant GCF investment criteria for the project with project indicators in the project design are given throughout this document.

It is important to note the importance of developing a project-specific Theory of Change to support the investment framework implementation and articulate how the project results will be achieved. The Theory of Change should be provided as a diagram illustrating the methodological approach in designing the project while linking project goals with outputs, outcomes and assumptions under which the project should be developed. More information regarding the development of the Theory of Change is available in GCF's Programming Manual.

1.4 Structure of this document

The GCF Water project design guidelines consist of two parts:

- Part 1 (the current document) provides generic practical guidelines and a stepwise approach for project design that meets GCF requirements
- Part 2 (complementary document) provides additional context and guidance for when applying the guidelines to specific water sub-sectors as identified by GCF: Integrated Water Resource Management (IWRM), Climate Resilient (CR) WASH, Drought Management and Flood Management.

This document, Part 1, is structured as follows: chapter 2 describes the main elements for a GCF project proposal design with specific emphasis on the climate rationale, before chapter 3 elaborates the structured stepwise approach for project design.

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2 Developing an impactful water programme or project for GCF financing

This document describes how to design a project in such a way that will become eligible for GCF funding. The activities needed for designing a good project can be substantial, depending on the size and complexity of the project. Studies needed to design a project can be financed by GCF's Project Preparation Facility.

This chapter describes in section 2.1 the key principles for developing an impactful water project eligible for GCF funding. Section 2.2 outlines the climate rationale needing to be included in the project proposal. This section also describes the recommended risk-based approach to be applied, in particular for projects focused on dealing with strong climate variability (such as drought and flood risk projects). Section 2.3 explains the link between the proposed project or programme with the country's NAP and NDC's. Finally, Section 2.4 describes how to use this guide.

2.1 Key principles for developing an impactful water sector project

Although each project is unique in scope and conditions, there are several key principles that need to be considered in developing a good water sector project:

- the project should be 'owned' by the stakeholders (country ownership)
- the project should take an integrated view on the water system, hydrological cycle and climate risk reduction; and
- project formulation should follow a structured process with key checks and balances.

2.1.1 Ownership of project – stakeholder involvement

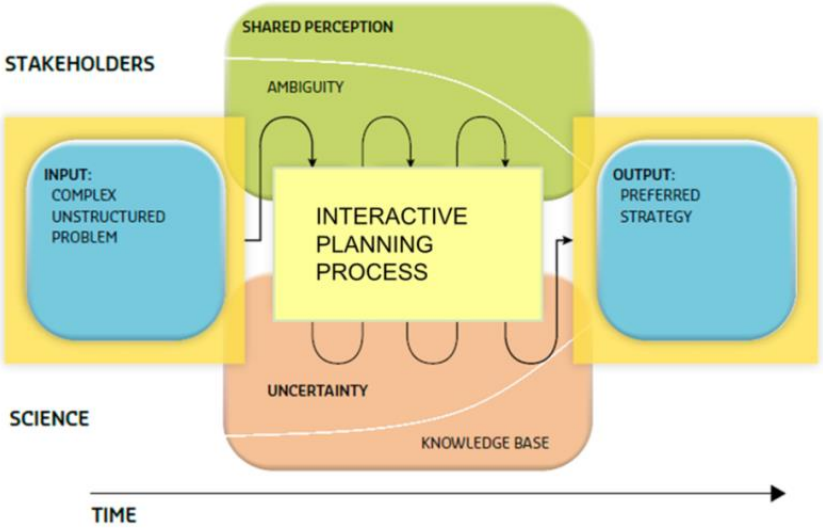
The first condition is that the project is owned by key stakeholders, i.e., the beneficiaries of the project as well as the implementing partners. For this a participatory approach in developing and implementing the project is needed. One of the first steps in the design of a project is to carry out a stakeholder mapping in which the beneficiaries are identified (and their wishes) as well as the implementing partners (and their mandates). The project design should be viewed as an interactive process of stakeholder engagement, that works towards a shared perception of the problems and solutions. In this process any ambiguity on what precisely are the problems, objectives and strategies is transformed into a shared perception on these topics and ultimately the formulated project. A process of joint fact finding with frequent interactions will also help to create a common knowledge base and a common view on the system to be studied. A shared perception of problems and potential solutions will also promote ownership of the project to be developed. Figure 5 illustrates this process.

Which stakeholders to involve depends on the problem and the project that is proposed. The many interactions between components in the water system (see section 2.1.2) causes that each intervention in the system (a project) can impact many people and interests, also outside the water sector. This includes:

- Other governmental agencies such as ministries or departments at the same governmental level (horizontal coordination) as well as higher or lower governmental levels (vertical coordination).
- Direct users (such as water user associations and the private sector), but also upstream and downstream users of the proposed project site.

Attention should be given to make the stakeholder process socially inclusive, improving the ability, opportunity, and dignity of those disadvantaged based on their identity (gender, age, minority groups, etc.). More information on stakeholder engagement will be given in section 3.3.2.

Figure 5: Engagement of stakeholders in project design

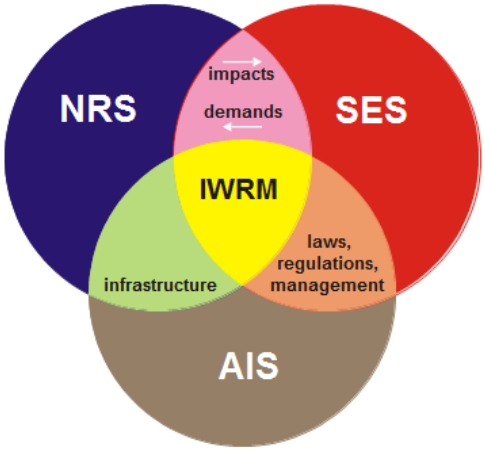


Involving the stakeholders in project design is not only needed to build stakeholder ownership of the proposed project. Stakeholders, and in particular the people that are dependent on the water system, have a lot of knowledge on that system, its problems and possible solutions. Making use of this (indigenous and local) knowledge will enhance the quality of the project design. Indigenous knowledge built through generations can play a significant role in facing contemporary challenges and may reveal improved solutions to climate adaptation and mitigation.

2.1.2 Integrated approach – take a systems perspective

The integrated character of water management requires that a good water project considers all interactions and cascading impacts that it may have with other components of the Water Resources System (WRS). Even if the project is developed for a specific purpose (such as a wastewater treatment plant), it must also consider the impacts the project will have on other system components. Such an integrated view on the WRS is illustrated in Figure 6 that identifies three sub-systems: the Natural Resource System (NRS), the Socio-Economic System (SES), and the Administrative and Institutional system (AIS). The figure also shows the relations that exist between these sub-systems. Considering all these sub-systems with their interrelationships is at the heart of Integrated Water Resource Management (IWRM).

Figure 6: Systems thinking - integrating the sub-systems NRS, SES and AIS



Source: Deltares (2017)[9]

[9] Loucks, D.P., van Beek, E. (2017). Project Planning: Putting It All Together. In: Water Resource Systems Planning and Management. Springer, Cham. https://doi.org/10.1007/978-3-319-44234-1_13

The **natural resource system (NRS)** consists of the streams, rivers, lakes and their embankments and bottoms, the groundwater aquifers, and the water itself. This includes the abiotic (or physical), biological, and chemical ('ABC') components in and above the soil. It also includes the infrastructure needed to collect, store, treat and transport water such as canals, reservoirs, dams, weirs, sluices, wells, pumping stations, pipes, sewers, and water and wastewater treatment plants, and the policies or rules for operating them.

The **socio-economic system (SES)** comprises the water-using and water-related human activities. This component can also include the stakeholders, i.e., the interested and affected public.

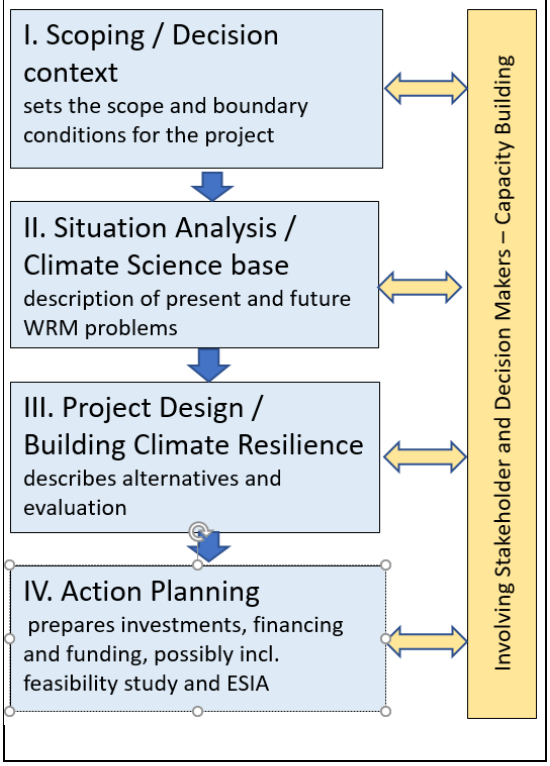
The **administrative and institutional system (AIS)** comprises the institutions that are responsible for the administration, legislation, and regulation of the supply (NRS) and demand (SES) components. The AIS includes all institutions that plan, build and operate the infrastructure required for water security. This includes both soft (regulations, policies, acts, laws, etc.) and hard (organisation, governance, structure, etc.) institutions, but may also include informal institutions such as community groups, volunteer networks, etc.

All these systems and their interactions should be considered during project design. The level of detail for their analysis depends on the specific project and how much it will interact with the different system components, i.e. the project's footprint or area of influence.

2.1.3 Structured Process in Project Design

The project proposal should provide specific information on the project for which funding is requested. This includes a description of its objective (i.e. the problem that it is trying to address) and socio-economic context. It should explain why that specific project is proposed and not an alternative. Moreover, it should describe how the project will be implemented. A specific condition for GCF projects is that the project should be designed to address problems caused by climate change (see section 2.2.1).

Figure 7: Structured approach for project identification and implementation



A common analytical approach for the development of a project or programme is given in Figure 7. It distinguishes four phases, each with its own focus. The first three phases prepare and design the project. The resulting 'project' is then translated into an action (investment) plan in the fourth phase. A generic description of this Framework for Analysis is given in Deltares, 2021[10].

The four phases are described in detail in the next chapter. Each phase describes a specific part of the project design process. Climate change and building climate resilience must receive specific attention in all phases, including:

- **Scoping/Decision context.** Setting the objectives and assessment indicators should demonstrate a clear focus on enhancing climate resilience or mitigation. If appropriate, a link with the National Adaptation Plan and the National Determined Contributions should be made.
- **Situation analysis/Climate science base.** During the scenario and/or risk assessment for the current and future situation (i.e., without taking actions) a range of climate change scenarios/projections must be analysed alongside socio-economic developments. By assessing these both individually and in combination, the additional contribution of climate change to any potential impacts can be assessed.

[10] Deltares, 2021. Framework for Analysis for water resources planning and water project preparation

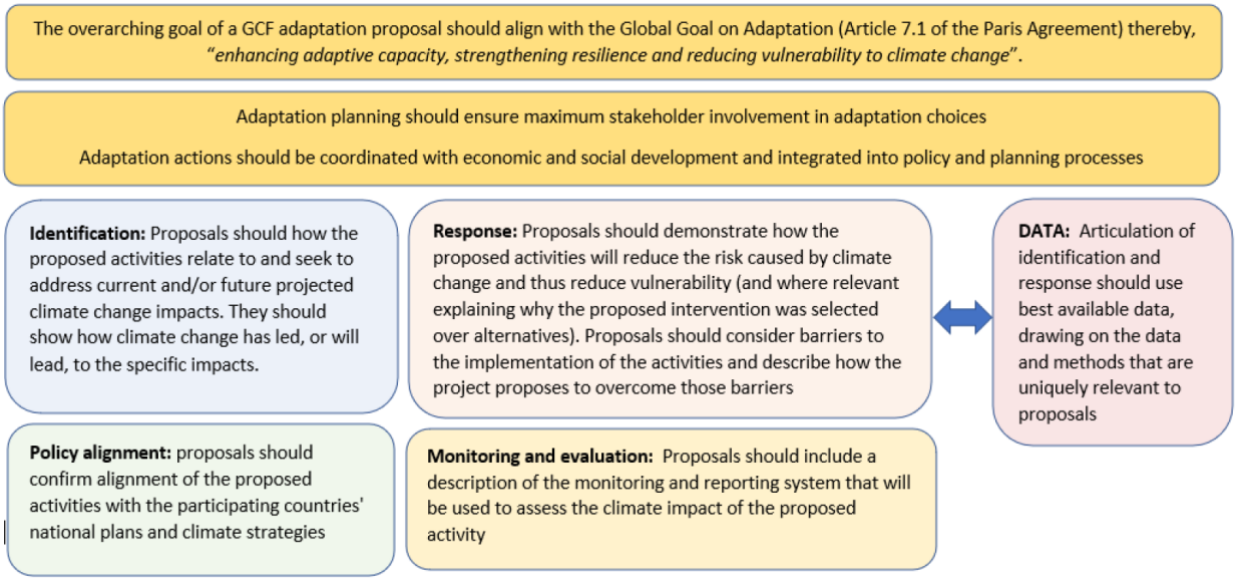
- **Project design.** Taking longer term climate change impacts into account during early stages of the analysis may result in the promotion of measures that are more flexible and/or robust to the uncertainties involved than 'traditional' BAU measures. Climate risk assessment will be key element to design adaptation pathway. Adaptation pathways, or sequenced strategies commencing with more flexible, low-regret options to which extra robustness may be added over time, may need to be developed. Often, measures yielding additional mitigation benefits will emerge as low-regret.
- **Action planning.** Traditionally, the water sector is mostly funded by public institutions or governmental organizations. Climate change might bring in other financial players including the private sector; for example, local businesses also have a growing interest in safeguarding their water supply under changing conditions. A co-investment approach opens new business cases for the implementation of measures.

2.2 Climate Rationale in project design

GCF is mandated to expand collective human action to specifically respond to climate change. For GCF funding, projects should response to issues induced or enhanced by the changing climate. This needs to be demonstrated by the climate rationale, as an essential integrated part of the entire project design to support a country's NDC and NAP.

2.2.1 Climate rationale for GCF funding

Figure 8: Conceptual framework for guidance, selection and assessment of a GCF adaptation proposal



Source: https://www.greenclimate.fund/sites/default/files/document/gcf-b33-05.pdf

The overarching goal of GCF adaptation proposals is that they align with the global goal on adaptation (Article 7.1 of the Paris Agreement) of "enhancing adaptive capacity, strengthening resilience and reducing vulnerability to climate change". Therefore, all proposals should in some way demonstrate how they achieve this or provide the "climate rationale" for the intended project. To create a better common understanding of how to formulate a climate rationale, GCF has published a policy paper[11]. This policy paper proposes that the inclusion of a climate rationale ensures that projects are designed to address climate change. It can be thought of as the overarching narrative that contributes to all GCF investment criteria (especially Impact Potential). For example, adaptation actions in the water sector should provide a well-evidenced analysis to demonstrate that

[11] https://www.greenclimate.fund/sites/default/files/document/gcf-b33-05.pdf

proposed activities are likely to be an effective adaptive response to the threat of specific climate change hazard(s).

Adaptation proposals therefore should contain the following elements (see Figure 6):

- **Identification**: Proposals should demonstrate how the proposed activities relate to and seek to address current and/or future projected climate change impacts. Proposals should identify the systems at risk and the climate change hazard affecting them. Under the 'Needs of the recipient' criterion, specific climate vulnerabilities of areas, people and water infrastructure can be identified that require urgent response.
- **Response**: Proposals should demonstrate how the proposed activities will reduce the risk caused by climate change and thereby reduce vulnerability. This element is related to the 'impact potential' criterion.
- **Alignment:** Proposals should confirm alignment of the proposed activities with the participating countries' national plans and climate strategies (e.g., NAPs. NAPAs, etc.). Projects should enhance country ownership and contribute to paradigm shifts for water security,
- **Monitoring & evaluation**: Proposals should include a description of the monitoring and reporting system that will be used to assess the climate impact of the proposed activity.

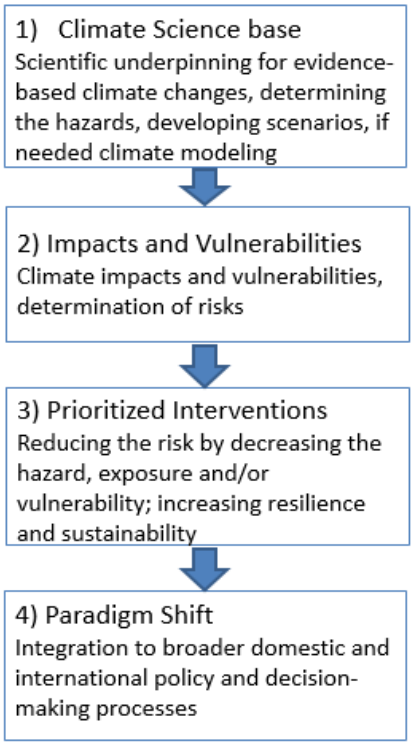
The climate rationale should be supported by the best available data, information, evidence, and knowledge. In the remainder of this report this is referred to as the **climate science base**.

The climate rationale in a GCF proposal provides the underpinning to ensure a specific project responds directly to climate change challenges.

To integrate the climate rationale properly in your project design, the following steps need to be taken:

1) Establish credible **climate science, base** underlining the climate hazards.
2) Robust assessment of exposure, impacts, vulnerability and disaster risks to **identify** priorities for adaptations and/or mitigations.
3) **Develop** a set of robust interventions that collectively and comprehensively address underlying climate risks and maximize sustainable development benefits.
4) **Align** interventions with the broader national and international policy and decision-making processes.

Figure 9: Climate rationale for water sector



These steps are illustrated in Figure 9. The first three steps are explained in a generic way in the next sections. The chapters on the sub-sectors (Part 2) will provide more specific information on these steps.

2.2.2 Climate Science base preparation

The climate science base is the evidence needed to convincingly demonstrate the need for the project to address urgent climate change risks beyond historic climate variability. A more extensive interpretation is that the climate science base should demonstrate to what extent the activities proposed in the project reduce these climate change risks and impacts.

The first activity in developing the climate science base is to conduct desktop research and literature review on available data and information on historic climate and expected climate change (temperature, rainfall, etc.) in the project area and the hazards associated with these changes (floods, landslides, water shortages, etc.). In most cases useful information will be available in existing documents on historic variability, hazards and climate change in the country. For further analysis it is obviously key to collect the available climate data. If possible, use should be made of climate observations and historical records from national meteorological institutes. Often these institutes have also prepared climate change scenarios for the country. However, often local observational data availability is limited in spatial, temporal resolution and coverage. In that case, global processed observational ground and remote sensing data sets and reanalysis of meteorological centres can also be of additional help. If climate projections are not available, they may also be obtained from gridded globally available data sets).

Climate related scenario elements are temperature, rainfall, evaporation, sea-level rise, and the occurrence of extreme events (such as droughts and typhoons). The parameters to be considered in the project design depend on the specific type of the project, e.g., evapotranspiration would be very important for a drought management project but not so much for a flood management project. Attention should be paid to inherent uncertainties in climate trends and variability both in past and future climate, such as precipitation decrease and seasonal and annual rainfall variability.

It is important to capture the key weather characteristics that cause malperformance. Historical observations therefore need to be verified if the length of climate records is sufficient to capture the long term and has a sufficient time resolution to capture short term variability of the climate indicator under consideration. For some projects this means that monthly average climate data are sufficient; for other projects it could mean that extreme values should be collected. The choice of which climate variables need to be collected or assembled depends on the choice of assessment indicators selected during Phase I of project design. Historic records are also often used to detect trends. The significance of those trends should be assessed before using the outcomes.

For future climate, the uncertainties can be expressed by a range of possible climate projections or scenarios[12].

Using climate projections or scenarios

Climate change modifies the characteristics of the historic climate. The extent and way these modifications may take place are projected by Global Circulation Models (GCMs) that simulate coupled atmospheric-oceanic processes determining the earth's climate. These GCM's are regularly being run and updated, among others to support the assessments of the IPCC under assumptions of different Greenhouse Gas (GHG) emission scenarios that describe global climate mitigation efforts. The most well-known global effort of standardization and intercomparison of the experiments carried out with these GCMs is the Coupled Model Intercomparison Project (CMIP5[1], CMIP6). From CMIP5 and the CMIP6, climate projections of all IPCC emissions scenarios (coupled to potential socio-economic pathways) are available for users world-wide. These climate projections can be used in multiple ways for water sector climate performance or risk assessment:

- Directly as input to water system / hazard models
- As a delta-change applied to the historic records that are used in the analysis. The deltas are obtained by calculating the difference between the baseline and future scenario run of the same model. Examples of such analyses can be found in Mendoza et al. (2018)[13].

[12] Climate projections and scenarios are often used interchangeably. Direct outputs from GCM's is most often termed a projection while climate scenarios often are based on selected model projections sometimes enriched with additional information.

[1] https://esgf-node.llnl.gov/projects/cmip5/

[13] Mendoza, G., A. Jeuken, J.H. Matthews, E. Stakhiv, J. Kucharski and K. Gilroy (2018). Climate Risk Informed Decision Analysis (CRIDA): collaborative water resources planning for an uncertain future.UNESCO, Paris. 162 pages ISBN:978-92-3-100287-8

The following considerations are important in the use of climate projections:

- The resolution (>100km grid size) of the models is very coarse for local applications. Dynamical downscaling (using Regional Climate Models, RCM) or statistical downscaling to local observation stations or grids can help to increase this resolution. The CORDEX[14] community is providing climate projections on a regional scale at higher resolutions (20-50km) for different world regions.
- The projections contain a bias compared to the observations. Deviations for temperature of 2 degrees are for example no exception. For this reason, local station or gridded observations are needed to correct the biases. Especially if the climate projections are to be used as direct input to water models, such bias correction is essential. Different portals nowadays are offering downscaled and bias-corrected GCM projections (e.g. GCF/WMO portal, WB Climate data portal[15], KNMI Climate explorer[16], ISIMIP[17]).
- The spread in projections between the different GCM's can be very high, even sometimes showing opposite trends. To use the projections properly to represent existing uncertainty, it is advised to always use multiple models and RCP's in the analysis. If a selection of GCMs needs to be made, the performance for the historical climate is often assessed to select the ones that are best capable the simulate this. Some countries may already have selected GCM or RCM projections for national use as part of their NAP efforts. In this case it is advised to use these data.
- The concept of confidence (IPCC, 2010[18]) can be used to assess the uncertainty in the projected future climate effects and impacts. E.g., if there are multiple projections agreeing on the sign and roughly size of the change and the projected trends are also in line with observed changes and scientifically plausible there also will be a high confidence in the projected climate effects.

2.2.3 Impacts, Vulnerabilities and Risk Assessment

Once a good overview has been obtained about possible climate change and related hazards, the next step is the determination of the possible impacts involved. To be able to demonstrate the contribution of the project in reducing the impacts of and increasing the resilience to climate change, the analysis can be decomposed into 4 components:

1. An analysis of impacts under historic climate variability to identify those climatic events that lead to undesirable performance or outcomes. Historical climate analysis therefore should also go hand in hand with historical analysis of hazards and reduced performance.
2. An analysis of impacts and vulnerability under future climate change to assess how the plausibility/likelihood of impacts may change and the size of impacts may develop.
3. An analysis of impacts and vulnerability under future socio-economic changes to assess the magnitude of any influence of other drivers besides climate change.
4. An analysis on to what extent implementation of adaptation measures (the project) is reducing current and potential future impacts to acceptable levels.

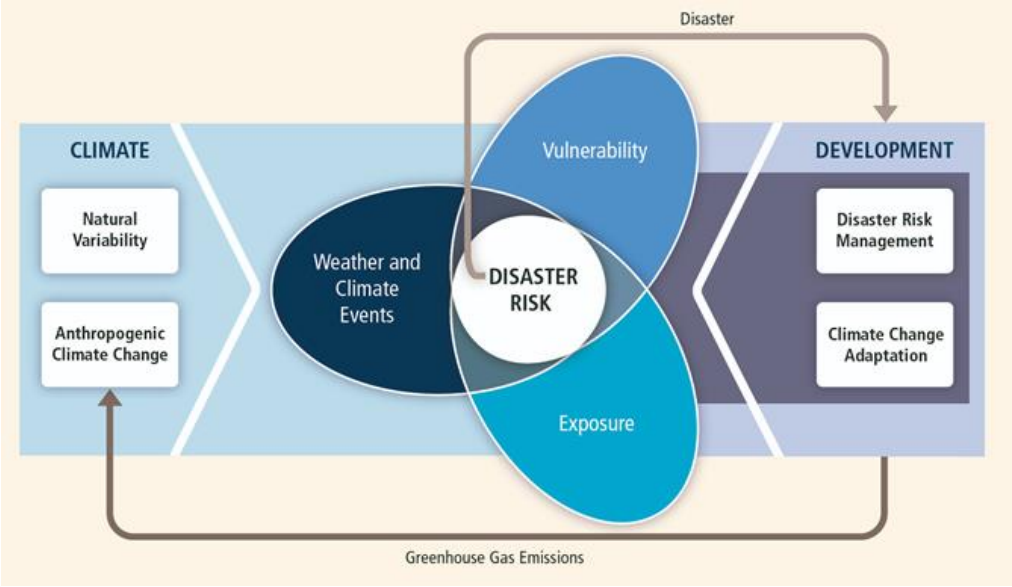
Climate change will increase the probability of hydro-climatic extremes such as droughts and floods. The risk involved is determined by the combination of the weather and climate events leading to hazards, and the exposure and the vulnerability of people, assets, ecosystems to them, as illustrated by the risk-based framework of IPCC for disaster risk (Figure 10). From an analytical point of view this framework can be expressed as risk = f (hazard, exposure, vulnerability).). Proposed projects can reduce the risk by influencing the hazard, and/or the exposure and/or the vulnerability. Consequence is a key component of assessing risk,

[14] https://cordex.org/
[15] https://climateknowledgeportal.worldbank.org/
[16] https://climexp.knmi.nl/start.cgi
[17] https://www.isimip.org/
[18] Mastrandrea, M.D. et al. 2010: Guidance Note for Lead Authors of the IPCC Fifth Assessment Report on Consistent Treatment of Uncertainties. Intergovernmental Panel on Climate Change (IPCC).

and a higher consequence from a hazard event or persistent stressor (e.g., seasonal shift in rainfall) exacerbated by climate change, will naturally lead to a higher risk rating. Exposure and vulnerability could also magnify the consequences/impact severity.

The risk assessment in project design will answer the question what the drivers for risk are and what the best pathways to reduce the risk are. At the same time, it will assess the impacts of the proposed project. These impacts can be either defined at full risk level (e.g., the expected damage over the range of return periods) or separately for the different components of risk, targeting specific vulnerability or exposure reduction objectives (e.g., number of citizens exposed to hazards). Risk assessments are particularly important for drought and flood management projects but should also be addressed in project proposal for the two other sub-sectors: IWRM and WASH. More information and recommendations on risk assessment for the sub-sectors will be given in Part 2 addressing these sub-sectors.

Figure 10: Risk Framework of IPCC



Source: IPCC, 2012[19]

Risk-informed decision-making on water-related risks

GCF is promoting a decision led approach to vulnerability and risk assessment, understanding the drivers of the risks and choosing the right metrics for the risk assessment. This implies that first the objectives of the main stakeholders involved should be established in a collaborative manner and that next the analysis should focus on those climatic and non-climatic factors that will pose the highest risk in achieving those objectives. The formulation of the objectives of the project is an important step in the structured process (to be further explained in section 3.3.4).

The planning process to support risk informed decision-making should look for a '*proportionate' response* to risk: it entails a transparent process of estimating the risk and of assessing the impacts (costs and benefits) of risk reduction and management. Planning for management of water-related risks should be guided by several core principles:

- Absolute protection from disasters (zero risk) is not feasible; hence the need for a search for a proportionate response to water-related risk.
- Decisions should be based on a thorough understanding of risk and uncertainty of both current and possible future water-related risks, including the impacts of climate change. Risk and uncertainty should be widely communicated.

[19] IPCC, 2012. Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation. Field, C.B.et all (Eds.) Cambridge University Press. https://www.ipcc.ch/report/managing-the-risks-of-extreme-events-and-disasters-to-advance-climate-change-adaptation/

- Plans should consciously evaluate a wide portfolio of responses. Measures selected for implementation should utilize resources effectively and fairly and should be flexible and robust.
- Plans should reflect the local context and be integrated with other planning processes. Responsibilities for governance and action should be clearly defined and communicated.

2.3 Linkages with National Adaptation Plans and NDCs

The proposed project will need to have a logical strong and consistent linkage with the National Adaptation Plan (NAP) and the National Determined Contributions (NDCs). That linkage can have two directions. The first direction is that the proposed project is part of the NAP as a concrete action. For GCF, this link is important as it indicates the ownership of the country over the proposed project. The link also provides evidence that the project is aligned with other strategic developments in the country.

The other direction is that the project contributes to make the NAP and NDCs more concrete in terms of programming and specification. While the need to address climate change through water is well recognized, this is not always being translated into concrete action in most NDCs and NAPs. In the NDCs submitted by countries as part of the Paris Agreement, water is, in many cases, recognized in terms of policy statements or broad strategies – usually in terms of adaptation. But few NDCs have included the intention to prepare a specific water portfolio. And while most countries acknowledge water in their NDC's 'portfolio of actions', few have estimated the related costs of these actions, even fewer have included detailed water-related project proposals. The project proposals as submitted to GCF can make this more concrete.

2.4 Applying these guidelines for project preparation

The key principles and climate science articulation described in this chapter and the structured process of project design that will be presented in the next chapter are quite extensive in the amount and level of detail on items to be addressed in project design. They describe an 'ideal' situation; following all these guidelines will be a challenge, particularly in situations with limited capacity and data. It will not always be possible (and needed) to address all items in detail. The guidelines should be seen as a checklist and depending on the specific project some elements will be fully addressed, while other elements may receive less attention or even may be omitted. In evaluating the proposals for funding, GCF will take this into account.

3 Structured Process in Project Design

Section 2.1.3 described the structured process to be followed in designing a project. Four phases were identified: I Scoping, II Situation Analysis, III Project Design, and IV Action Planning. This chapter will present the details of these phases. First a description will be given of the terminology that will be used in this chapter (section 3.1). Next the generic structure and the steps to be taken during the four phases are given in sections 3.2-3.6. Making use of that (traditional and local) knowledge will enhance the quality of the project design.

3.1 Definitions used in the structured approach

The main terms that this guide is using in the structured approach are highlighted in Box 1. An overview of the terminology and definitions as used in these guidelines is included in the glossary.

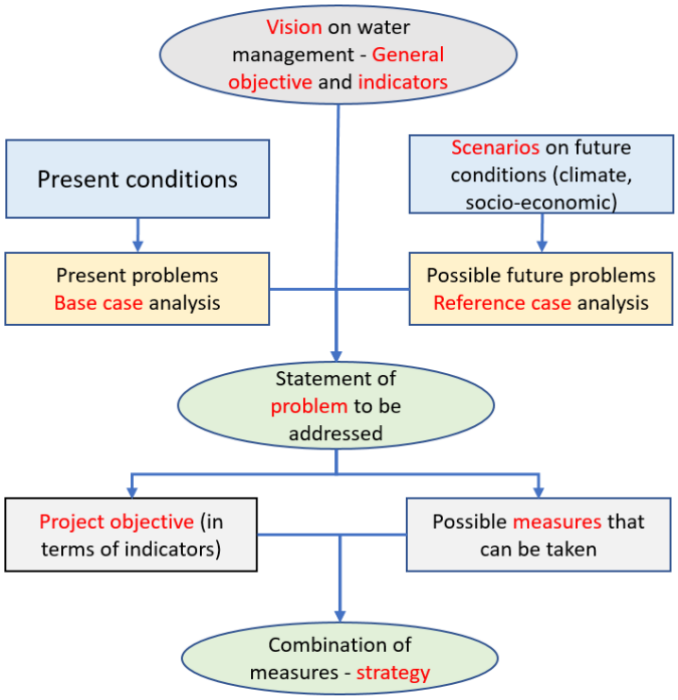
The basic analysis approach for developing a project and the terminology involved is illustrated in

Figure 11.

Box 1: Definitions

Objective: what do we want to accomplish?
Strategy: how do we want to do it?
Measure: what are we going to do?
Pathway: what is the sequence of measures or actions to achieve the objective?
Scenario: the external economic, environmental, or political situation affecting the impact of your strategy or project.

Figure 11: Approach and terminology in developing a project



The structure of the approach is as follows:

- The approach commences with a **vision** that the government has on climate-resilient water management and the services that depend on the water resources. This vision is operationalized by

formulating a **general development objective** for climate resilient WRM and services and defining **indicators** that will express in how far this objective is reached.

- Next the **problem** in the water system should be identified, provoked and/or exacerbated by existing climate variability (a drought, insufficient safe water supply, etc.). The present problem is called the **base-case**.
- In solving the present problem, the problem owner (e.g., a governmental agency) wants also to prevent potential future problems, e.g., because of increased population and/or climate change – for this they must develop **scenarios** on possible futures. The future situation without new measures is called the **reference case**.
- The combination of the present problems and possible future problems are combined in a project specific **problem statement** that the proposed project must address.
- The problem owner defines how the preferred improved situation should look like – the **project objective**. The project objective is specified by setting targets for the indicators as defined above.
- Alternative **measures** (interventions) are proposed to solve the present and future problems. A comparison of these alternatives and their economic benefits, environmental and social impacts and risk analysis will be presented to decision makers.
- A particular combination of measures is the **project** (or project portfolio) that the problem owner wants to adopt to solve the present problem and prevent future problems.

3.2 Introducing the main steps for project development

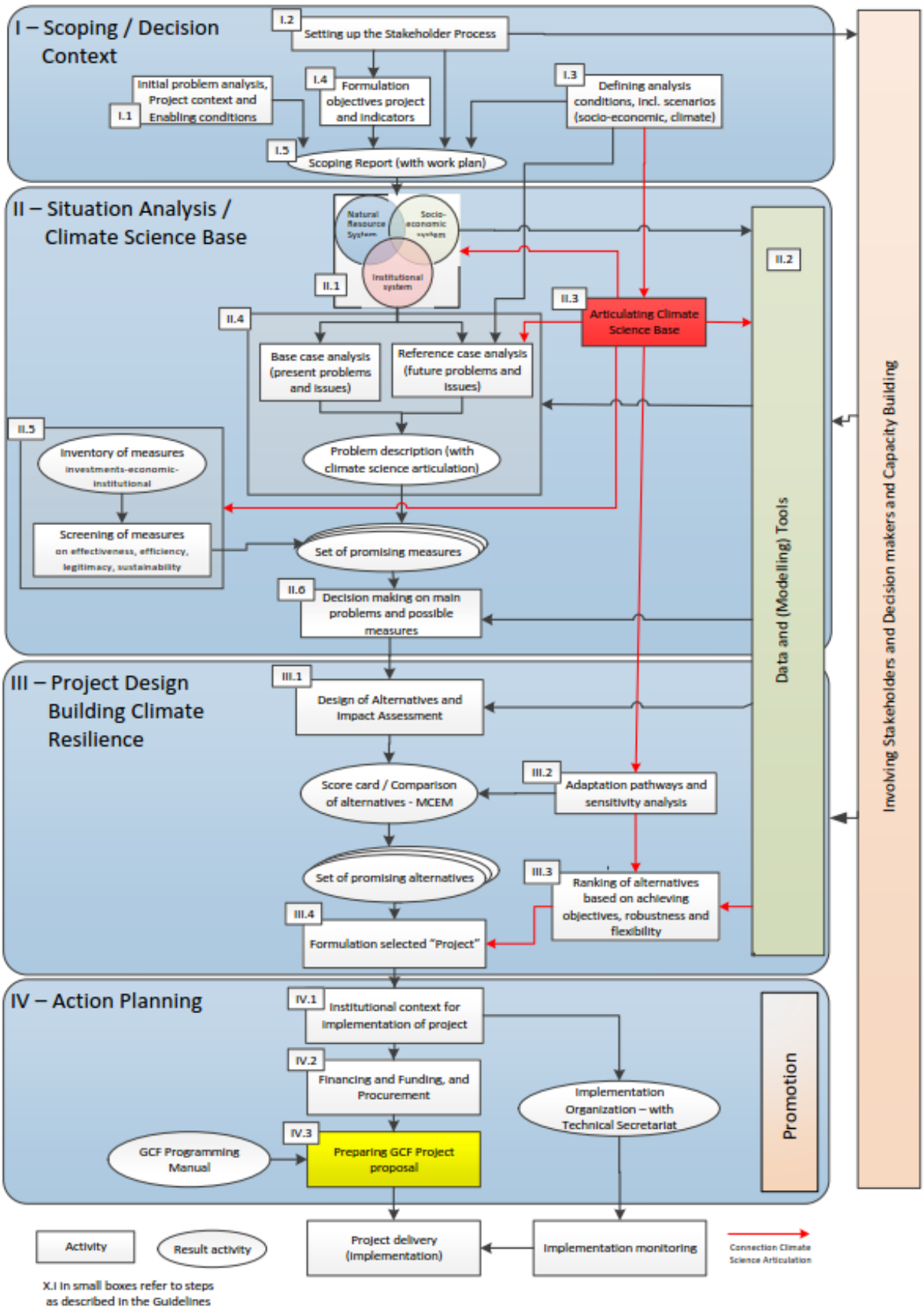
A structured approach for the development of a project or a set of projects towards climate resilience is presented in Figure 12. It distinguishes 4 phases, each with its own focus. The first 3 phases are studying the problems, developing the climate science base and designing a climate resilient project. The outcome will be a proposed project or a set of projects (programme). In the 4th phase the project or programme will be made 'bankable[20]' by including a specification of its feasibility, detailed environmental and social impacts, corresponding financial and funding arrangements as well as a description of the implementing arrangements. The numbers in Figure 12 refer to the steps that will be described in the next sections.

These guidelines are for carrying out a study to design the 'project' that will be proposed to GCF. Projects can have many forms and sizes. The level of detail to which the project proposal must pay to address the phases of Figure 12 will depend on the type of project. A project might consist of a combination of several measures. The words 'measures', 'interventions', 'actions' as used in these Guidelines are synonymous and are used to identify individual project components.

The following sections are set up as an example 'study' to design a project. It describes the activities that should be addressed in the study. In the actual project proposal to GCF only the results of the activities will be included as was illustrated in Figure 4.

[20] In this context 'bankable' is defined as a project or proposal that has sufficient collateral, future cashflow, and high probability of success, to be acceptable to institutional lenders for financing (World Bank definition).

Figure 12: Structured approach for project development and implementation



For instance, for each phase of project development, an analysis of relevant investment criteria must be conducted. Project proponents must assess the relevance of all six investment criteria and indicators to demonstrate project alignment with GCF's investment framework. Since the final project should be aligned

against GCF's investment criteria, one criterion can be considered in more than one phase and across steps of project development, with different perspectives and levels of information to inform the final assessment.

Note that for each investment criteria different aspects will be taken into consideration regarding country and project contexts. For more details, please refer to GCF's investment criteria mentioned in chapter 1.

In addition, for the development of the concept note, proponents should consider the six investment criteria according to the GCF concept note users guide[21]. Pre-feasibility studies, and economic and financial justification for the grant elements and their level of concessionality[22] should be included, however, the level of detail of those documents are not the same as for the funding proposal.

For the funding proposal, a more in-depth description of the project, climate context, implementation arrangements and Theory of Change narrative are necessary. The expected performance against the six investment criteria should again be described. For specific aspects of developing a funding proposal, GCF funding proposal template guidance note should be followed.[23]

The investment criteria should provide consistency and transparency in funding proposal documents, increasing the efficiency of proposal assessment. Besides, the criteria indicators are used on an individual project context, therefore, a project can be more related to one criterion than others. The application of indicators are country and context specific.

Table 1 describes how each investment criterion is related to each phase and step of project design, at both concept note and funding proposal levels. Further explanation about the components of the concept note and funding proposal can be found in the GCF Programming Manual.

Sections 3.3 to 3.5 explain in more depth each phase of the project design and steps involved to produce the final project proposal.

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[21] gcf-concept-note-user-s-guide.pdf (greenclimate.fund)

[22] As per Board decision B.07/06, concessionality in GCF is widely understood as "fundingwith below-market terms and conditions" where the minimum amount of concessional funding needed can be up to and including the full cost of the project or programme. Concessionality can also be defined as a measure of the level of benefit provided to a borrower when compared with financing available at full market rates. Concessionality aims at lowering the cost of borrowing or minimizing the risk in a transaction for the borrower. GCF offers concessionality in order to facilitate a high-impact climate action that would otherwise not take place.

[23] fp-template-guidance-note.pdf (greenclimate.fund)

Table 1: Investment criteria against project design phases and steps on concept note and funding proposal levels

Investment criteria	Information needed	Phase	Step	Concept note	Funding proposal
Impact Potential	Formulate objectives and indicators	Phase I - Scoping Phase - The Decision Context	Step I.3 Defining the analysis conditions, including climate change challenges	Describe the climate vulnerabilities and impacts, GHG emissions profile, and mitigation.	Describe project benefits for mitigation and/or adaptation, and how the project contributes to sustainable development pathways. Calculations necessary.
			Step I.4 Formulating the project objectives and climate resilience goal		
	Develop climate scenarios assessing current and future vulnerability. Identify problems and possible interventions.	Phase II - Situation Analysis including climate rationale	Step II.4 Problem analysis	Environmental and social risk screening	Environmental and social impact assessment
			Step II.5 Identification and Screening of Potential Measures		
Paradigm shift	Determine the paradigm shift related to the project implementation	Phase I - Scoping Phase - The Decision Context	Step I.4 Formulating the project objectives and climate resilience goal	Estimation of expected impact aligned with paradigm shift potential.	Contribution to paradigm shift the project aims to support. Scale, replicability and sustainability dimensions to be considered.
		Phase III - Project Design - Building Climate Resilience	Step III.4 Formulation of the project		
Sustainable development potential	Economic, social and environmental co-benefits related to the project	Phase III - Project Design - Building Climate Resilience	Step III.3 Evaluation and ranking of alternatives and decision making	Brief description of how project intends to benefit target population	Outcome mapping to GCF results areas and co-benefit categorization
Needs of the recipient	Identify and analyse financial, economic, social and institutional needs and barriers that prevent achieving the project objective	Phase I - Scoping Phase - The Decision Context	Step I.3 Defining the analysis conditions, including climate change challenges	Describe main needs and barriers that need to be addressed and how the project aims to address them. No need for specific indicators	Core enabling environment indicators should be added to reinforce needs of the recipient. The scale of vulnerability and how the projects aim to address the issues should be included.
		Phase II - Situation Analysis including climate rationale	Step II.4 Problem analysis		
	Identify how the project will address the context specific needs and barriers	Phase III - Project Design - Building Climate Resilience	Step III.4 Formulation of the project		
Country ownership	Identify alignment with NDCs relevant national plans indicator, and/or enabling policy and institutional frameworks.	Phase I - Scoping Phase - The Decision Context	Step I.3 Defining the analysis conditions, including climate change challenges	Description of how project fits country's priorities	Policy context and plans part of the administrative and institutional framework should be considered
	Stakeholder engagement process, including national authorities	Phase I - Scoping Phase - The Decision Context	Step I.2 Setting up the stakeholder engagement process	Describe engagement among relevant stakeholders that took place and plan to develop funding proposal	Describe how the country takes ownership of and implements the project. Add links with existing climate strategy, alignment with policies and frameworks other GCF projects, capacity to deliver, engagement with stakeholders.

Efficiency and Effectiveness	Co-benefits and monetization coupled with efficiency and effectiveness assessment of alternatives.	Phase III - Project Design - Building Climate Resilience	Step III.3 Evaluation and ranking of alternatives and decision making	Justify rationale and level of concessionality, minimum requirements and most efficient plan for viable investment	Describe efficiency and effectiveness of project against its mitigation and/or adaptation objective. Include indicators and reinforce the need for GCF's funding for the project to take place.
			Step III.4 Formulation of the project		
	Institutional arrangements and financing	Phase IV - Action Planning, Financing and Implementation Arrangements	Step IV.1 Institutional context for implementing the project		

3.3 Phase I - Scoping Phase – The Decision Context

This scoping phase identifies the content and context of the project (what is included within the scope of the project and under what conditions), its objectives (desired results of the project), and any constraints (boundary conditions and limitations). Based on this analysis, intensive communication with decision-makers is required on the approach to the remainder of the analysis (the next phases).

The main elements of the Scoping phase are:

1. Initial problem analysis, project context and enabling conditions
2. Setting up the stakeholder engagement process
3. Defining the analysis conditions, including the climate change challenges
4. Formulating the project objectives and climate resilience goals
5. Decision-making: Scoping report with work plan

The result of the Scoping phase can be presented in a scoping report, which includes the work plan for the remaining phases of the project preparation process.

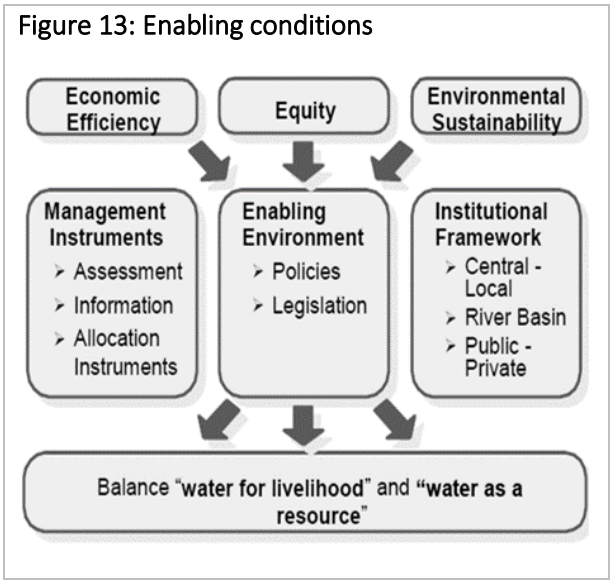
3.3.1 Step I.1 Initial problem analysis, project context and enabling conditions

The first step is to carry out an initial problem analysis, describing the context of the envisioned project as well as analysing the conditions needed to successfully implement the project.

In the initial project preparation analysis, a first inventory of the current and expected future problems and possible impacts on the socio-economic system and the environment must be undertaken. This initial project preparation analysis will determine what is the context of the problem the project is trying to address, and which stakeholders will need to be involved in project preparation.

The context gives a first introduction in why the envisioned project is needed and the potential institutional setting for the project that might be required. In terms of development objectives, links should be made with national laws, policies, strategies and plans including the NAP, NDC's, NDA's, etc. It should also present some climatological evidence for the need of the project, based on historical events and/or research. Further details of these items will be developed during the next steps in the study, especially in Phase II on the Situation Analysis.

Figure 13: Enabling conditions



Source: Adopted from GWP, 2005, Integrated Water Resources Management Plans – Training Manual and Operational Guide, GWP, Stockholm.

Projects can only be successfully carried out if a minimum system of institutional and technical conditions are in place. Most of the enabling conditions are external to the activities of the project but will greatly determine its success in implementation. In practice, implementation arrangements are generally defined early in the project preparation stage. A generic description of the enabling conditions for integrated planning is given in GWP (2000)[24]. These enabling conditions are illustrated in Figure 13. During the Scoping phase it should be determined which conditions are relevant for the specific project. This will also depend on the climate resilient issues involved. Institutional strengthening measures can be included in the project to be developed.

3.3.2 Step I.2 Setting-up the stakeholder involvement process

The next step is to set-up the stakeholder involvement process. In general, two categories of stakeholders can be identified:

- the people and organizations that will be affected by the project; and
- the people and organizations that are needed to implement the project.

Membership to each of these categories should be according to social inclusiveness, with particular attention given to vulnerable groups on account of race, gender, citizenship, ethnicity, disability and economic status. The second category will play a major role in Phase IV in which it will be decided which organization(s) will be responsible for the implementation of the Project and how the project will be funded.

From a climate change point of view, it is important that institutions dealing with climate science as well as NGOs are represented in the stakeholder process and are consulted. NDA's are also key stakeholders, as they are the developers and gatekeepers of the GCF country program, in which all intended GCF-funded projects for the country will be included. One should also ensure the meaningful representation from the groups that are most vulnerable to climate change in the intervention area.

In most cases it is advised to carry out at the start of a project a stakeholder mapping to determine which stakeholders should be involved. Reference is made to various manuals (e.g., CDI[25], World Bank[26], Schmeer[27], ODI[28]) on how to perform stakeholder mapping and to set-up a stakeholder process.

3.3.3 Step I.3 Defining analysis conditions and climate induced risks

In addition to the more legal and institutional conditions as described in the previous section it is necessary to obtain agreement on the boundary conditions of the technical content for the proposed project, including:

- The **base year** that will be used in analysing the project; this is often the most recent year for which basic data on the present situation is available.
- The **time horizon**(s) of the project:
    - May include short term (e.g., 5 years), medium term (e.g., 20 years) and long term (>40 years) horizons; climate change considerations demand inclusion of a long term time horizon;
- The **discount rate** to be applied in the economic analysis:
    - taken as specified by (e.g.) the Ministry of Finance or Economic Affairs, or by other financiers of the planned investments (e.g., ADB, World Bank, JICA);
- **System boundaries** of NRS, SES and AIS - the components and the level of detail that will be included, for example:

[24] GWP, 2000. Integrated Water Resource Management. TEC Background Papers No. 4. Global Water Partnership. Stockholm

[25] CDI (2016) The MSP Guide, How to design and facilitate multi-stakeholder partnerships, Practical Action Publishing Ltd, UK

[26] http://www1.worldbank.org/publicsector/anticorrupt/PoliticalEconomy/stakeholderanalysis.htm

[27] Schmeer, Kammi. 1999. Guidelines for Conducting a Stakeholder Analysis. Bethesda, MD: Partnerships for Health Reform, Abt Associates Inc.

[28] https://www.odi.org/publications/5257-planning-tools-stakeholder-analysis

- Will impacts on the coastal zone be included in a river basin project?
- Are results to be presented at local government unit level?

- **Scenario** assumptions concerning factors external to the WRS, such as climate change, the growth of population, food and energy consumption and prices.

The choice of the **time horizon** is often given insufficient attention. Official planning horizons (e.g., five, ten and twenty-five years) are typically used as time horizons for elements of the analysis. However, one should also consider the time scales of the system and the processes within it. The most important timescale involved is related to the lifespan of the project (the amortization[29] period of the investment). Investments that will be susceptible to longer-term impacts will need to use projections of the climatological conditions over a longer period (e.g., at least 50 years), even if the planning horizon of the project itself is only 25 years.

A good project should address both present and potential future problems. To predict possible futures, **scenario assumptions** must be made on possible socio-economic developments (e.g., population growth and future economic activities) and on climate change (including sea-level rise, increase in rainfall, etc.). An economic evaluation of a project may require assumptions about the future prices of energy and food. It is important that the project proposal is clear about the assumptions made. From the particular perspective of GCF, sufficient attention must be given to the climate scenarios.

A basic approach to developing project scenarios is to adopt national assumptions on socio-economic development and climate change as found in formal governmental policy documents. Economic scenarios often include high-low (growth) or optimistic-pessimistic scenarios. A Business-As-Usual (BAU) scenario should also be defined that assumes that present trends will continue. The various scenario components can be combined into a set of different views on how the future may develop.

3.3.4 Step I.4 Formulate project objective and climate resilience goal

An essential activity in the scoping phase is the translation of general development objectives (as described in policy documents like NAPs and NDCs) into an operational objective for the project that can be quantified. What goals and objectives are connected to the needs for climate resilience of the project? What evaluation metrics and indicators are appropriate?

The primary objective of a GCF funding proposal is to enhance the climate resilience outcomes from the proposed project. However, this should complement the socio-economic development objectives of the project. This guidance suggests commencing with the socio-economic objectives and once set, supplementing these with the climate resilience goals. In this a link to the Investment Criteria of GCF should be made. This is explained further below.

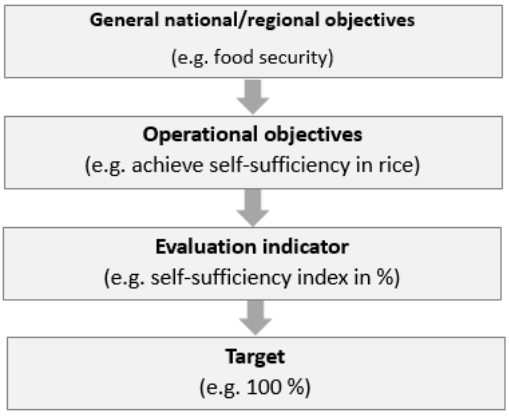
National and regional development objectives

An essential component of a project that is to be well integrated with other components of the WRS is the connection of the project and its objectives to national development goals and common international goals (e.g., the Sustainable Development Goals - SDGs). The project should refer to national policy priorities and indicate the contribution the project will make to the various development goals. Required information is usually described in various national policy documents. In addition to these, any existing regional / provincial policy documents must also be considered. A project must have an agreed objective that not only focuses on the main purpose of the project, but also expresses its relation to national and other sector plans, as well as its contribution to realizing these plans.

[29] the action or process of gradually writing off the initial cost of an asset.

Operational objectives, evaluation indicators, and targets

Figure 14: Making objectives operational



Policy documents often specify objectives in general terms. These general objectives must be translated into operational objectives preferably in socio-economic terms, which are more meaningful for decision makers and stakeholders. Evaluation indicators should be defined for each objective as a measure to show how far these objectives will be achieved. If possible, clear targets should be defined. This process is illustrated in Figure 14.

The evaluation indicators need to be comprehensive (i.e., sufficiently indicative of the degree to which the objective is achieved) and measurable. The indicators do not all have to be expressed in a single measurement scale. Indicators can also be expressed in monetary and non-monetary terms. It is essential to incorporate sustainability as a key component to the project objectives, preferably relating this to the UN Sustainable Development Goals (SDGs) and relevant SDG targets and indicators.

Climate resilience and adaptation goals

In addition to the 'general' project goals as described above, specific attention is needed to also include climate resilience outcomes in the list of project objectives. This is typically achieved through a reduction in either the exposure or vulnerability to the impacts of climate change and provides the climate rationale for a GCF project. In including relevant climate resilience objectives, the following should be considered:

- Uncertainty in climate change, natural variability, exposure, and vulnerability. In practice this means that not one single project investment can be defined to guarantee sufficient additional resilience. Additional objectives stemming from decision making under uncertainty can be helpful here such as whether the investment generates extra **robustness** for the water resources/risk management system against acceptable costs and multiple additional benefits (low regret), or the **flexibility** of the investments to be expanded over time with growing insight into climate change, natural variability, exposure, and vulnerability.
- In low- and middle-income country contexts, performance objectives for water security are often set higher than what has been achieved under historical climate conditions. In fact, there are two adaptation components: one to adapt the system to current climate variability and one to adapt it to future climate variability under climate change.

3.3.5 Step I.5 Decision-making - Scoping Report with Work Plan

Once it is clear 'what' and 'what not' will be included in the project and 'why', the project developers must specify 'how' this will be done. This requires that the following subjects are addressed:

- The project objectives and the indicators to express and assess how far the objective is achieved. These are described above.
- The activities for the next phase and their level of detail. Care should be taken that the level of detail to which the activities will be carried out are at an appropriate level: not too detailed and not too limited. The level of detail will be determined by the issue at hand and the objective of the project.
- The computational tools that will be used and data availability. Data availability determines which computational tools can be used. If necessary, monitoring programs should be developed to gather data. The issue at hand determines which computational tools should be used; e.g., for IWRM, WASH and drought projects usually a water balance model is needed and for flood projects a hydrodynamic model. These will be further specified in the next phase.

- The way stakeholders will be involved in the project. It is strongly advised to include the stakeholders also in the analytical work, to develop ownership of the analysis results.

The results of the scoping phase are documented in a scoping report. This report can serve as a reference during the execution of the project preparation. An essential part of the report is the proposed work plan, in which time, budget and human resource allocations are needed for project preparation. This work plan typically includes bar charts (possibly derived from critical path analyses) for activities and staffing, time schedules for deliverables, milestones, reporting procedures and similar features. The report should include a communication plan that describes the planned interactions between the decision-makers and stakeholders and the project preparation team.

The scoping report is more than a workplan. It should contain all that has been learned in the Scoping phase and that has been agreed between the project developer, decision-makers, and stakeholders. The scoping report must be formally accepted by the stakeholders as the basis document for activities in the next phase.

3.4 Phase II - Situation Analysis including climate rationale

In this second phase the water system is described and analysed. Where possible a quantitative analysis will be carried out using data and models. Scenarios are developed that describe possible future situations. Special attention is given in this Phase to the proper selection of plausible climate boundary conditions for the assessment of current and future climate vulnerability and the development of climate robust and flexible interventions that reduce this vulnerability. Present and possible future problems in the water system are identified and possible interventions are assessed to address these problems. This will be done from two perspectives: i) water as a resource and accompanying water services, needed for human consumption and economic activities, and ii) water as a risk to be managed. Depending on the type of project two approaches may have to be used in the analysis: i) carrying out a systems analysis approach (see for example Figure 16), and ii) carrying out a risk analysis. Although both approaches are very valuable, often for managing water risks only the second approach is applied. All activities must be carried out in close interaction with stakeholders. From a climate change point of view, it is important in this Phase to define the quantifiable climate rationale behind the proposed project.

The main steps in the Situation Analysis phase are:

1. Understanding and describing the water resource system.
2. Data and Modelling Tools: Collecting data and developing quantitative tools and models on the system.
3. Develop the science base for the Climate Rational (Climate science base).
4. Problem analysis: Formulating a quantified problem statement for both present and possible future problems.
5. Identification and Screening of Potential Measures: Providing an overview of possible interventions to solve these problems.
6. Decision making on main problems and possible interventions.

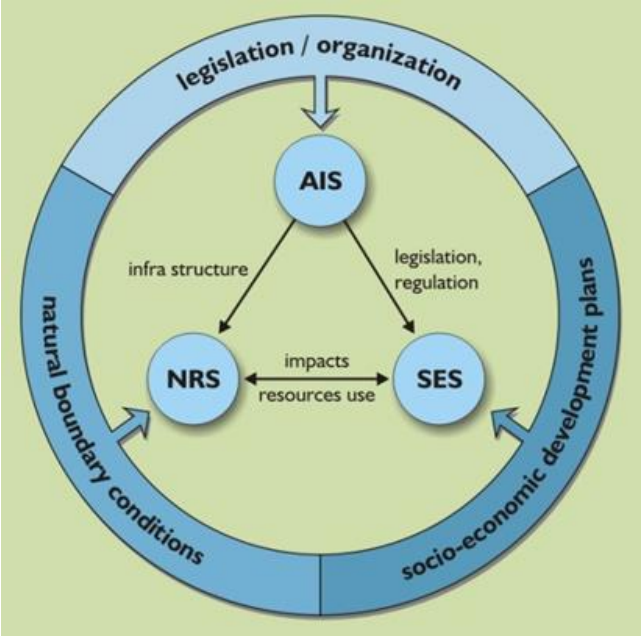
The ultimate result of the Situation Analysis phase should be discussed with the stakeholders, resulting in decisions on the focus of the next phase, in particular what the main problems are to be addressed and the most promising interventions to solve these problems.

3.4.1 Step II.1 Understanding and describing the Water Resources System

The first step is to get a good understanding of the water resources system in which the proposed project will function. As explained in section 2.1.2 a WRS comprises the Natural (Resources) System (NRS), the Socio-Economic System (SES); and the Administrative and Institutional System (AIS). Each of the three systems is embedded within its own environment. The Natural Resources System is bounded by climate and (geo)physical conditions. The Socio-Economic System is formed by the demographic, social and economic conditions of the

surrounding economies. The Administrative and Institutional System is formed and bounded by the constitutional, legal, and political system. Figure 15 shows the systems embedded in the overall environment.

Figure 15: The Water Resources System and its context



Source: Loucks, D.P. and van Beek, E., 2017a, Water Resources Systems Planning and Management, an Introduction to Methods, Models, and Applications. Springer Publishing.

It is important that the project development process includes a good description of the integrated elements of the WRS. Most decision-makers and stakeholders will be non-technical or only know about a limited part of the overall system. To be able to make balanced decisions they should understand how the overall system functions and how interventions in one part of the system will impact other systems elements.

Natural Resources System (NRS)

The NRS comprises the natural and engineered infrastructure, including the hydro-meteorological boundary conditions. Models can be used to simulate the processes of water flow distribution through the system, considering the storage of water in infrastructure, natural waterways and the soil and water withdrawals including evapotranspiration to satisfy the demands of water-using activities. Analysing the NRS also provides the information of the hazard and exposure as needed for the risk analysis (see section 2.2.3).

Socio-Economic System (SES)

Analysing the SES includes the demands that the SES expects from the NRS (quantity and quality), the impacts of the SES on the NRS (such as pollution), as well as the impacts that the conditions of the NRS has on the SES (quality, flooding, shortage, etc.). For the SES also models can be used to quantify these demands and impacts. The SES analysis provides information on the vulnerability of the system to changing conditions. For GCF funding proposals, specific attention must be given to the impacts that climate change might have on people, livelihoods, and how those impacts are distributed.

Administrative and Institutional System (AIS)

An analysis of the AIS is required to identify any legal, regulatory, or institutional constraints on climate-resilient water resources management. Attention must be given to the interaction between various authorities involved in climate-resilient water resources management such as water resources management, disaster risk management, agriculture, environment, and hydrometeorology and to the effectiveness of their interactions. For GCF funding proposals an assessment must be made in how far the existing AIS can enable climate-resilient

water resources management and how the institutional framework can be strengthened to deliver better results. The AIS analysis is mainly descriptive and does not include modelling.[30]

In describing the above systems, specific attention needs to be given to those components that will be influenced by climate change. This includes water supply and demand (the water balance) as well as the risk elements (climate variability) for floods and droughts. The knowledge gathered (preferable translated in models) should enable the assessment of the impacts of climate change in socio-economic terms. The analysis of the AIS can result in recommendations for institutional changes to enable an investment environment for water resources management.

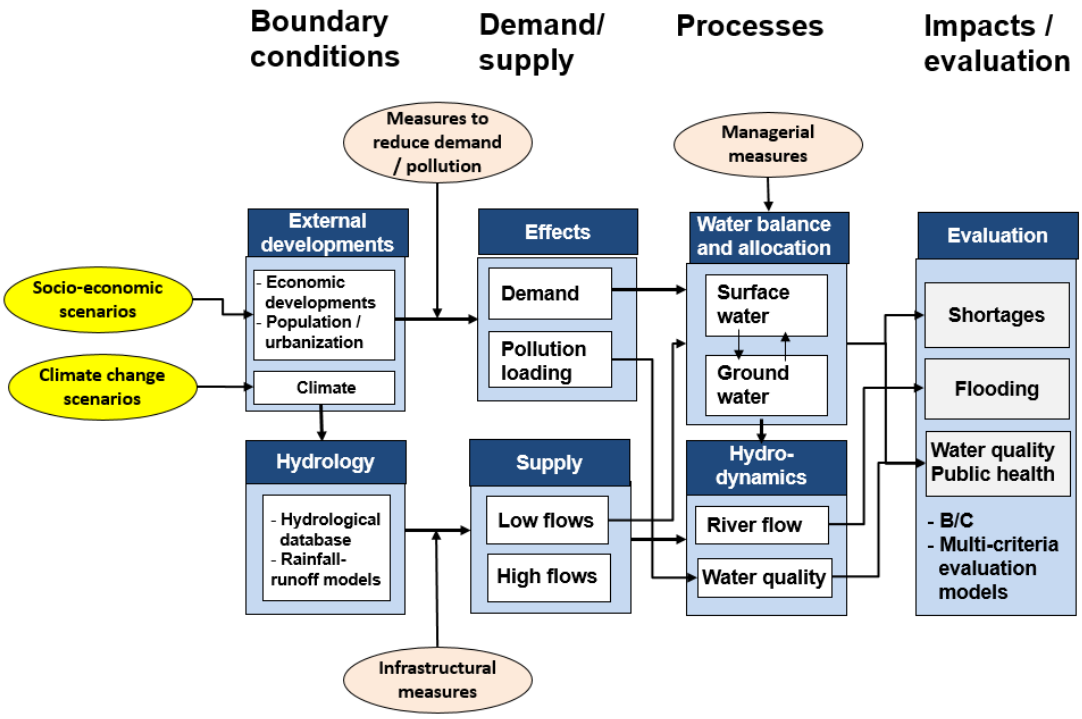
3.4.2 Step II.2 Data and Modelling Tools (computational framework)

The result of the data collection and modelling activities is a quantitative representation of the WRS and the SES at an appropriate level of detail. The modelling tools enable assessment of the effects of individual measures or combinations of measures, expressed in values for the evaluation indicators chosen. Applying computer models may require considerable effort. Careful selection of the phenomena to be represented by the models, tuned to the needs of the project, is important.

IWRM projects can use integrated modelling to quantify the impacts of decisions across the triple bottom lines (environment, social and economic) and how to better develop and manage water resources systems.

The kind of models that are required to develop a project or program depends on the specific conditions of the project involved. Drought projects require hydrological / water balance types of models. Flood projects require hydraulic modelling. WASH projects can be supported by both water quantity and water quality models, since the impact of climate change on WASH services leads to issues related to both water quantity and water quality. An illustrative schematization of integrated models is given in Figure 16, including how alternative interventions and different scenarios can be analysed with this framework.

Figure 16: Illustrative integrated modelling approach for water resource projects



[30] The EPIC Response Framework provides a systematic approach to assess the evolution and the interaction of institutions involved in climate-resilient water resources management.

The goal of using data and modelling tools is to enable informed decision making on the proposed projects or plan. To get ownership of the results of the models it is strongly advised to follow a collaborative approach in the analytical work with the data and models. Stakeholders have often a good knowledge of certain components and involving them in the analytical work will not only result in such ownership but also increase the quality of the data and models.

The model system should enable the quantification of the present situation (the base case, see section 3.1) as well as predicting the impacts of interventions (measures) and future developments (scenarios). As illustrated in Figure 16 the model system will provide also input in the screening process of measures (step II.5), the decision making on the main problems and promising measures (step II.6), the impact assessment of alternatives (step III.1) and decision making on the final project to be proposed (step III.4).

3.4.3 Step II.3 Articulating the climate science base as input for the Climate Rationale

As mentioned in the previous chapter, a clear climate rationale is required. For this the approach can be followed as described in Section 2.2 (and illustrated in Figure 9) and use can be made of the various Climate platforms available (and referred to in 2.2.2). The Climate Rationale is started here in Step II.3 of the Structured Approach but climate rationale-related components are addressed in several other Phases and Steps of the Structured Approach as presented in Figure 12 (between brackets the terminology used by GCF in Figure 9):

- Climate Science Base (Data)
  - Analysis of historic data on water availability, hazards in relation to historical weather is done in Step II.1 (analysis of NRS) and Step II.2 (data and modelling).
  - Developing climate scenarios and determination of hazards; partly this is done in Step I.3 (analysis conditions) and further analysed in Step II.2 (NRS and SES under a changed climate); if needed this will include climate modelling (GCM and RCMs).
- Impacts and Vulnerabilities (Identification)
  - Identification of the stakeholders is done in Step I.2 while the analysis of their climate resilience objectives and indicators (metrics) is done in Step I.4.
  - The impacts and vulnerabilities are determined in the analysis of the SES system in Step II.1 and further explored in the problem analysis in Step II.4, supported by the model and data system as developed in Step II.2.
  - The risk assessment is carried out in the problem analysis of Step II.4 for the present and future situations, including the situation under climate change.
- Prioritized Interventions (Response)
  - The inventory and screening of possible interventions is done in Step II.5.
  - The comparative analysis of (alternative) interventions is done in Step III.1 and includes a multi-criteria evaluation.
  - The prioritization may include an adaptation pathway approach and sensitivity analysis in Step III.3 to deal with the uncertainties involved in climate change.
  - Decision making on the selected intervention (project or set of projects) is done in Step III.4.
- Paradigm Shift
  - The desired paradigm shift will be addressed in Step I.4 in formulating the objective of the project. Indirectly this is also addressed (to some extent) in determining the institutional context for the implementation of the project.

3.4.4 Step II.4 Problem analysis

Problems in the water sector will in general have two dimensions: a systems dimension and a risk dimension. For the problem analysis the data and modelling tools are used for:

- The definition of the present problems – the **base case** analysis
- The definition of possible future problems – the **reference case** analysis

In a next step the impacts of alternative interventions will be determined. This will be done **in reference** to the problem situation that includes possible future problems due to socio-economic developments and climate change. The impacts of these interventions will be compared to the **reference case(s)**, explaining the name.

Systems perspective

Problems related to the systems perspective have a focus on dealing with average conditions: to provide sufficient supply to meet the needs of the SES, to ensure that the water quality is acceptable, etc. IWRM and WASH projects usually address the systems perspective. Important elements are the water balance (demand versus supply) and water quality (mass balance of pollutants).

Risk perspective

The risk perspective has a focus on the extreme events and is in particular relevant for projects with a strong Disaster Risk Management (DRM) orientation such as floods and drought management. The risks involved include present risk (due to climate variability) and future risk (in which climate change will change climate parameters). The risk approach includes defining the (probability of the) event, the exposure, and the vulnerability, as explained in section 2.2.3. The problem can be formulated at full risk level (e.g., expected damage with a certain return period) or separately for the different components of risk targeting specific vulnerability or exposure reduction objectives (e.g., number of citizens exposed to hazards, coping capacity built among a number of organizations).

It is noted that all projects will have systems and risk dimensions. Also, in 'systems' projects it will be needed to include the risks associated with climate variability and climate change. The difference is in the attention that need to be given to the risk dimension. The analysis of risk will be very prominent in flood and drought projects and will get less attention in the case of IWRM and WASH projects.

Quantified Problem Statement

In the Scoping phase the objective (what we want to achieve) and indicators (how we are going to measure our performance) has been defined. The problem statement should give an overview of the 'scores' of the indicators for typically 3 situations:

- The present situation (the base case)
- The future situations (the reference case – possibly for several scenarios)
- The desired situation (the targets)

The presentation of these situations is usually done in a score card. The use of score cards will be explained in Section 3.5.3. The problem statement is reflected in the first 3 columns of the score card. To be eligible for GCF funding it is advised to identify the problems caused by climate change separately from the problems caused by socio-economic developments, showing the additionality of climate change.

Barriers and Opportunities

In addition to the quantified problem statement as presented above, an overview needs to be given of the barriers that (might) prevent achieving the objectives. This is part of investment criteria 4 (needs of the recipient) analysis. These barriers can be:

- Enabling conditions barriers (see section 3.3.1).
- Financial barriers (financial viability of investments, markets, etc.).
- Environmental barriers (physical, geographical, etc.).
- Societal barriers (awareness, social norms, political influences, etc.).

Reference is made to the Water Security Sectoral Guide for a detailed description of potential barriers. Opportunities to overcome these barriers should be identified.

3.4.5 Step II.5 Identification and Screening of Potential Measures

Once the present and possible future problems have been defined, measures can be identified that will address these problems. An inventory should be made of all the measures that the stakeholders are proposing or considering or those that are included in existing development or management plans. Based on the quantified problem analysis additional measures might be formulated. The computational framework can be used to determine the environmental, social and economic impacts of these measures as well as in terms of risk management. The most promising measures will be kept for detailed analysis in the next phase in which the actual project is designed. For GCF projects specific attention should be given to climate change adaptation and mitigation measures that make the water resource system less vulnerable to climate change.

Measures can be divided into different categories: technical (investment), managerial, economic, ecological, and institutional. Examples of possible measures will be given in the chapters 5 to 8 on IWRM, WASH and Drought and Flood Management. An inventory of all possible kinds of actions that can be taken will in general result in a long list of possibilities. In most cases it will not be practicable to analyse all of these in detail.

A screening process is needed to select the most promising ones. Screening is a rapid and global evaluation of possible actions in comparison with a more thorough evaluation under step III.3 (see section 3.5.3). This can be done in several ways. It can be done by using the modelling framework developed for the project, but it can also be done by limiting the analysis to a few indicators, such as economic benefit or environmental or social impact. Best practice also suggests including an indication of the robustness and the flexibility of the proposed measures under the range of climate change uncertainty. Another kind of screening analysis is to apply expert judgment to evaluation criteria in line with the GCF investment criteria (see also section 1.3). Evaluation of project activities is also necessary to assess the **expected performance against investment criteria** (concept note part B.3 and funding proposal part D). In addition, the level of concessionality[31] for the project needs to be evaluated, i.e., the project needs to extend the minimum level of concessionality needed to make it viable and ensure the sustainability of measures. The stakeholders should be involved in this screening process. Box 2 highlights the evaluation criteria used for the evaluation of project activities in a project financed by GCF as an example.

[31] GCF Policy on Concessionality, 7 June 2021. https://www.greenclimate.fund/sites/default/files/document/gcf-b29-inf11.pdf

Box 2 Example of project under implementation and the evaluation criteria for the evaluation of proposed investments as part of the funding proposal. (https://www.greenclimate.fund/project/fp192)

The GCF has approved a climate resilience project for wastewater systems in Barbados with the goal to facilitate the enhancement of the health, wellbeing, and productivity of Barbadians through the use of carbon neutral and climate resilient water and energy management, technologies and strategies that ensures water is protected, managed, recycled, reused, and conserved.

Barbados is facing absolute water scarcity fostered by climate change-related threats such as flash-flooding events disrupting the sewerage system and a skewed pattern of precipitation and increased evapotranspiration which sets the stage to drought events. The project falls under the priority group Small Island Developing States and focus on wastewater management strategies using reclaimed water to help mitigate the impacts of climate change on water scarcity. In addition, one of its outcomes is to support farmers by replenishing aquifers that support irrigation developments, communities through increased employment opportunities, local businesses, the general population and the tourism sector through general water management improvements.

The funding proposal provided economic and financial justification for concessionality by highlighting the criticality of the relative contribution of GCF to the overall programme to realize its financial and economic benefits. It demonstrated strong impact potential by explaining how low-emission and climate resilient development likely are to achieve mitigation and adaptation impacts.
In addition, it addresses the paradigm shift criteria by enabling a shift in the wastewater management practices of Barbados that includes the use of low carbon and climate resilient water and energy management technologies. Furthermore, project scalability and replicability are fostered by a combination of activities that promotes private sector engagement and its innovative nature.

Moreover, the project's sustainable development potential is described through its contributions to SDGs 2, 4, 5, 6, 7, 9, 11, 13, 17. Therefore, contributing to the achievement of the Agenda 2030 and fostering environmental, economic and social co-benefits. In addition, there is a gender-sensitive campaign development to ensure stakeholders are aware of the impact of water quality and quantity availability on women and youth livelihoods. The recipient's needs criteria is addressed by an in-depth analysis of the country's vulnerabilities and environmental and social-economic risks due to adverse impacts of climate change and natural hazards that disproportionally affects youth and women. Country ownership is shown through the development of new laws and processes to adapt the country to the projected impact of climate change, such as the Barbados Growth and Development Strategy 2013-2030 and Barbados' National Climate Change Policy. Finally, the efficiency and effectiveness of adaptation and mitigation measures were showcased by thoroughly explaining the rationale for the grant financial instrument and co-financing by the BWA, the use of best practices and available technologies. Moreover, the project's legitimacy was ensured due to its reliability in robust legal and institutional frameworks and the engagement of the public sector, private sector and civil society during the project implementation. This is a role model that can be replicable in other countries and regions to treat water as new asset class.

Identification of no-regrets measures

A special category of promising measures is the 'no-regrets'. More realistically, we should speak of 'likely no-regrets' and 'low-regret' measures. These are measures that are expected to perform equally well under a range of future climate scenarios and that do not limit the application of other measures in the future. It is relatively straightforward to reach agreement among decision makers and stakeholders that these should be implemented, preferably as soon as possible, as these measures do not disadvantage any stakeholder. It should be confirmed that these measures will not have negative impacts on other measures or prevent other measures from being implemented in the future.

3.4.6 Step II.6 Decision making on main problems and possible interventions

This phase will result in a better understanding of the water resource system and in the formulation of the problems that the proposed project aims to address. It will also produce a set of promising measures that will be analysed in detail in the next phase. All these results should be documented in a report that must be accepted by all stakeholders as constituting the basis for the next phase. If stakeholders have different views these alternative views should be documented, and an explanation must be given on why these alternative views have not been taken into account. As much as possible the stakeholders must agree on:

- The functioning of the water resource system – does it contain the components that need to be considered?
- The problem statement – are these indeed the (priority) problems that the project should address?
- The possible interventions – does this overview include all (important) solutions that the stakeholders have in mind to address the problem? This might include the improvement, expansion or replication of on-going efforts based on stakeholder experiences and expert advice.

In other words, the content of this report should be shared with key stakeholders and their views and recommendations incorporated where needed.

3.5 Phase III – Project Design – Building Climate Resilience

In this third phase the project is formulated. The start is the set of promising measures from the previous phase. These measures are combined into alternative projects that are evaluated on their impacts to achieve the objective as defined in phase I. Specific attention should be given to the uncertainties associated with climate change and socio-economic development and for this an adaptive management analysis must be carried out. This provides the input for the evaluation of alternatives in terms of their robustness and flexibility to deal with these uncertainties. Based on this evaluation, the project can be formulated.

The main elements of this phase are:

1. Design of alternative strategies and impact assessment
2. Dealing with uncertainty - adaptive management analysis
3. Evaluation and ranking of alternatives and decision making
4. Formulation of the project - components and activities

3.5.1 Step III.1 Design of alternative projects and impact assessment

The design of a project or project portfolio involves the development of coherent combinations of promising measures to contribute to the achievement of the development objectives as formulated in Phase I and meet the targets as much as possible. As there are generally many indicators related to these objectives, and probably many of these objectives are expressed in different units, the design of projects is not a simple process. Relationships among combinations of measures and their impacts on the evaluation indicators are complex. The optimum combination may depend on who is asked. Trade-offs among the values of different indicators, and disagreements among various stakeholders, are inevitable. It should also be considered that the implementation time of measures will differ.

The design of a project or project portfolio is therefore an iterative process. One approach is to start by developing a project based on one of the more important objectives and develop alternatives around that first project design. These alternatives are defined by the boundaries of the solution space. Comparison of the economic benefits and the strategic environmental and social impacts of these alternatives can then lead to the construction of a compromise project by varying elements in the project. A resulting loss with respect to one criterion can then be compared with gains to another.

Climate resilient water security project design is an 'art' rather than a science and requires good insight into the functioning of the water system, the socio-economic requirements and impacts, and stakeholders' preferences. Project selection is often the result of negotiated interests and reflections on different views on how development should take place. The data and modelling tools can help determine the expected economic benefits and strategic environmental and social impacts of the project in terms of the defined indicators. In case models are not available, the impacts can be estimated by experts. These estimates must be confirmed and accepted by the stakeholders and decision makers in workshops.

The measures considered in this phase of the analysis are mostly at pre-feasibility level and can be added to the concept note formulation. In Phase IV - Action Planning, the selected measures will be worked out in more detail, including possible design and feasibility studies. In that phase also an ESIA (Environmental and Social Impact Assessment) might need to be carried out, which is necessary for developing a robust funding proposal.

As climate change raises threats of harm to human health and the environment, the **precautionary principle** should be applied in project design and preventive actions should be taken, even if some cause-effect relationships are not yet fully scientifically proven. According to the precautionary principle, uncertainty about the damage to be incurred should not serve as an argument to delay action. In the face of great uncertainty, a precautionary approach might result in a more stringent adaptation response.

3.5.2 Step III.2 Dealing with uncertainty - adaptive management analysis

The ultimate challenge is to make better informed decisions in water management under future uncertainties. Methods are available to make sure that the decision-making results into consistent, replicable, and accessible outcomes. The CRIDA (Climate Risk Informed Decision Analysis) approach might be employed to prepare such informed decisions[32]. CRIDA is an approach that implements decision scaling and bottom-up vulnerability approaches through collaborative stepwise planning procedures and Adaptation Pathways. The CRIDA approach incorporates both the Decision Tree Framework and Dynamic Adaptation Policy Pathways.

The Decision Tree Framework[33] is a repeatable process for the evaluation of climate change risks to new development projects. At the conclusion of this process, the project planner will be empowered to confidently communicate the method by which the vulnerabilities of the project have been assessed, and how the adjustments that were made (if any were necessary) improved the project's feasibility and profitability. The framework adopts a 'bottom-up' approach to risk assessment that aims at a thorough understanding of a project's vulnerabilities to climate change in the context of other non-climate uncertainties (for example, economic, environmental, demographic, or political). It helps to identify projects that perform well across a wide range of potential future climate conditions, as opposed to seeking solutions that are optimal in expected conditions but fragile to conditions deviating from the expected.

The Strategic Framework for Climate Resilient WASH Development[34] advances sector thinking around WASH and climate change. The Framework is grounded in a core objective: to provide sustainable WASH service delivery, both now and into the future. The emphasis is on climate resilient development, including strengthening the resilience of WASH systems and on investments to manage current climate variability, as well as long-term changes in climate. This encompasses both development and emergency preparedness

[32] UNESCO and ICIWaRM (2018). Climate Risk Informed Decision Analysis (CRIDA), Collaborative Water Resources for an Uncertain Future.

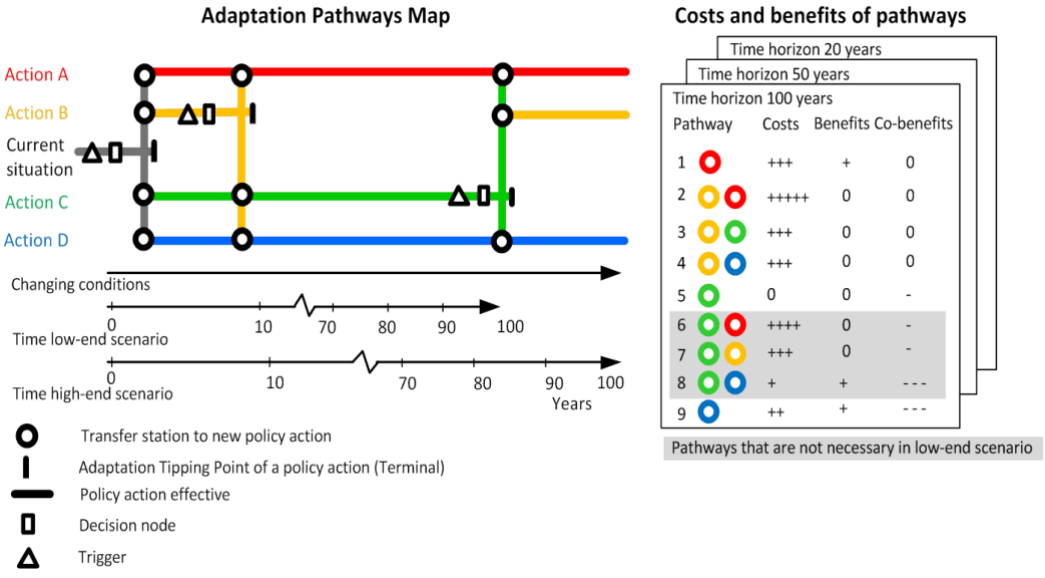
[33] Ray, Patrick A.; Brown, Casey M. 2015. Confronting Climate Uncertainty in Water Resources Planning and Project Design: The Decision Tree Framework. Washington, DC: World Bank. https://openknowledge.worldbank.org/handle/10986/22544

[34] https://www.gwp.org/en/WashClimateResilience/

programmatic spheres with climate resilience addressed as a cross-cutting issue encompassing elements of both disaster risk reduction (DRR) and climate change adaptation. The principles and practices outlined in the Framework aim to complement and strengthen ongoing national and sectoral adaptation planning processes, for example those under the National Adaptation Plans (NAPs) process where WASH sector needs are a key component.

The Dynamic Adaptative Policy Pathways (DAPP) approach[35] identifies tipping points that determine in time when a certain policy or intervention is no longer acceptable and (another) intervention is needed. Figure 17 illustrates the pathway approach. By exploring all possible interventions (actions in the figure) one can develop adaptation pathways that will minimize the regret. An adaptive plan specifies actions to be taken immediately to be prepared for the near future and actions to keep options open for additional adaptation if needed in the future. The exploration of adaptation pathways is one of the main ingredients of an adaptive plan. A monitoring system collects information to get early warning signals (triggers) for implementation of actions or for reassessment of the plan. The DAPP approach is illustrated in Figure 17 with at the left side an overview of possible pathways and to the right a (very simplified) 'score-card' to help evaluate the options.

Figure 17: Following an adaptive pathways approach



Source: Haasnoot, M., Kwakkel, J.H., Walker, W.E. and Maat, J.T., 2013, Dynamic adaptive policy pathways: A method for crafting robust decisions for a deeply uncertain world. Global Environmental Change. Vol. 23, Issue 2. Pp. 485-498

In preparing for uncertain future conditions, we do not want to take the risk that large investments are being made in the short-term which later appear to have been unnecessary. Therefore, two additional indicators are used in the evaluation: robustness, and flexibility; and a category of no-regret measures identified. In this evaluation, robustness and flexibility are taken to mean the following:

- Robustness: how well is the existing project performing when the future develops differently than expected? Will the project or project portfolio then still achieve the objectives?
- Flexibility: how adaptable is the project when it appears that the future develops differently than expected and we need to change the project?

[35] Haasnoot et al, 2013.

Robustness and flexibility often have a strong relationship with costs. A robust project can often be more costly (big reservoirs, high dikes, etc.). A flexible project (e.g., many small reservoirs, built over time) can also appear to be more expensive in the end. Cost comparisons need to be considered when deciding on a project.

Applying a DAPP approach requires resources in data, time and manpower. Different levels of analysis can be distinguished in terms of methods and techniques to be applied depending on the need of the case. Which one to use can be determined by answering the following questions:

- Should future uncertainties be included in the analysis?
- Is it beneficial to use the adaptation pathways approach?
- Which economic evaluation method under uncertainty should be used?

These questions are not easy to answer and the methods on how to deal with uncertainty in project evaluation are still developing. The need for these methods is growing, given the high uncertainty of climate change and the large impacts involved. For projects that require large investments (such as the construction of reservoirs) an adaptive pathway approach is strongly recommended. On the other hand, it should be recognised that these methods are very useful for countries with advanced management systems and access to a lot of reliable data. If these are not available projects should be assessed and prioritized through a multi-criteria assessment that includes criteria such as 'robustness' and 'flexibility'.

3.5.3 Step III.3 Evaluation and ranking of alternative projects and decision-making

Presentation of the selected promising projects to decision-makers may be done by means of briefings, presentations, and summary reports. The level of detail and the way the results are presented should give an overview of the results at an appropriate level of detail for the audience involved. Visual aids such as score cards and interactive computer presentations of study results are often very helpful for promoting a discussion of the results of the analysis.

The most straightforward way to present the impacts of selected alternative projects or strategies is done in matrix form by means of a 'scorecard'. The columns of the scorecard represent the alternative cases examined in the analysis. The rows represent the impacts of different alternatives with respect to the chosen indicators. An example of a scorecard for an IWRM project is depicted in Table 2. This example shows the impacts of two alternative projects in terms of the objectives and indicators as defined in Phase I. The Reference Case (predicting the scores of the indicators if no action is taken) makes a distinction between the case without climate and with climate change. Making this distinction will support the funding request for GCF funding. Scorecards can include quantitative results (in terms of money, %, etc.), and qualitative assessments by means of a relative score (e.g., classes or plusses and minuses), or by means of colouring (red is bad, green is good). The purpose of scorecard presentations is to present a visual picture of the relative attractiveness of the alternatives based on the identified indicators. The presentation of the results in scorecards allows a decision-makers to give each impact the weight they consider most appropriate.

Table 2: Example of a scorecard for an IWRM project, showing the scores of two alternative projects on evaluation indicators

Objectives and indicators	unit	Base Year 2020	Target 2040	Ref. case (no action) no CC 2040	Ref. case (no action) with CC 2040	Alternative projects Project 1 2040	Alternative projects Project 2 2040
Obj.1: Water and Sanitation							
% people access to safe drinking water	%	50%	73%	63%	63%	65%	73%
% people access to sanitation facilities	%	30%	70%	50%	50%	55%	70%
Obj.2: Food production							
Irrigation area	1000 ha	24	35	26	24	28	35
# animal water points	#	300	900	400	280	500	900
Obj.3: Industry and Energy							
Water supplied to mining	%	30%	90%	40%	25%	50%	90%
Water supplied to industry	%	70%	90%	70%	60%	80%	90%
Hydropower generated	MWh	34	120	34	28	70	120
Obj.4: Environment							
Protected watershed area	km^2	1200	3500	2000	2000	2500	3500
Number of springs/sources protected	#	300	900	400	400	500	850
Average class water quality rivers	I - V	II	IV	II	II	III	IV
Obj.5: Vulnerability							
Vulnerability to floods - average damage	m€/yr	120	< 50	100	200	100	50
Vulnerability to droughts - average damage	m€/yr	200	< 30	160	250	80	30
Implementation information							
Required investments	m€		-			400	1200
B/C ratio economic categories (Obj.2, Obj.3)	-		> 1,2			1.3	1.1

Source: Loucks, D.P., van Beek, E. (2017). Project Planning: Putting It All Together. In: Water Resource Systems Planning and Management. Springer, Cham. https://doi.org/10.1007/978-3-319-44234-1_13

The scorecard provides the information based upon which stakeholders can express their preference for a certain project or project portfolio and responsible officials decide which one they want to implement. This is not a straightforward process as the various stakeholders and decision makers will often have different opinions about priorities. The ultimate decision is often political and is more intuitive than analytical. Still, the quantified information made available by means of the scorecard will inform and influence these decisions.

Tools are available that help to 'rank' the alternatives. These so-called Multi-Criteria Evaluation Methods (MCEM) use standardization methods and weighting to come up with an overall score for each alternative. An issue with these methods is that the outcome depends on the methods chosen. An even more important problem with MCEM is that it is often very difficult to agree on the weights to be given to the indicators. For this reason, a negotiated ranking as a result of a joint decision process based on scorecards is often the best approach. Some of the widely utilized criteria to rank and select measures include those around the economic benefits, environmental impacts, social impacts, flexibility, robustness, and implementation feasibility (e.g., financial readiness or implementation arrangements). As part of the MCEM analysis a SWOT analysis can be carried out. The SWOT identifies the strengths, weaknesses, opportunities and threats of each project. Strengths and weaknesses are related to internal factors of the project that can make it more likely to respectively succeed or fail. Whereas opportunities and threats are related to external factors that could help or harm project implementation. A SWOT analysis is a tool to be used to identify opportunities for project improvements.

A cost-benefit analysis (CBA) is another potential tool available to help 'rank' alternatives. It is a technique that considers monetary and non-monetary measures and helps decision-makers to design or choose the best alternatives within a project plan. It determines how well or not a project may turn out. A social component can be added to the CBA, which is important because it comprises not just the financial effects of a project but also the social impacts such as pollution, drought, flood, employment rates, legal aspects, and many other impacts. It is a tool to investigate the strong and weak points of alternatives, and to select the most viable projects, considering their socio-economic impacts. The analysis relies on summing positive factors and subtracting negative ones to determine a net result. When a project's costs are less than the benefits derived from it, it is considered positive and viable. When the benefits derived do not compensate the planned financial investments, it is considered financially non-viable and negative. A CBA is a more complex, more data

intensive, analysis to evaluate and rank alternatives within a project and can be used to enhance the quality of funding proposals.

3.5.4 Step III.4 Formulation of project or project portfolio

Phase III finalizes the design part of the analysis. The results of the 3 phases and the final result (the project or project portfolio) is compiled in a project impact and design document. This document will be the base for the next phase: the action planning. It gives a comprehensive picture of the impact of the project on the integrated development and management of the water resource system, addressing the issues involved.

It should be an easily accessible piece of documentation for the stakeholders and should contain ample illustrative information (infographics). A suggestion for the general structure of such document is the following:

1. Introduction (why, how, objectives, participatory approach followed).
2. Description of the WRS (NRS, SES and AIS).
3. The policy context (institutional setting, general goals and policies, legal framework).
4. The analysis carried out (objectives and indicators used, scenarios, problems, measures).
5. Selected project (alternatives, adaptive approach, sensitivity analysis).

All technical detail can be included in Annexes and separate supporting reports.

The (technical) formulation of the project concludes the first three phases in which the project is designed from a content point of view. Once agreement has been reached by the decision makers on the content of the project, the institutional arrangements and financing must be addressed. That will be done in Phase IV.

3.6 Phase IV – Action Planning, Financing and Implementation Arrangements

In the Action Planning phase, the "Project" will be prepared for implementation. An implementation plan will be developed that describes for each project component what will be done, by who, how it will be financed, etc. Finally, institutional arrangements will have to be made to ensure a smooth implementation.

The main elements of this phase are:

1. Institutional context for implementing the project
2. Financing and funding of the project
3. Formulate the project in terms of the GCF Programming Manual

3.6.1 Step IV.1 Institutional context and implementation arrangement

The proposed project can consist of several components. Not all components have to be implemented by the accredited entity. In broader IWRM-type projects different institutions will be involved such as the Ministry of Housing for WASH projects and the Ministry of Water for river training projects. In some cases, the project will be implemented jointly by institutions from central, regional or local governments. It is considered good practice to have a single project implementation leader that is responsible for the implementation of the project (i.e. accredited entity) and is responsible to lead the collaboration with the other actors involved (executed entities and partners). This section is meant to provide guidance in case the project consists of several components that are being implemented by several implementing/executing entities.

The implementation of the approved project will be done on an investment or 'transaction' level. For this reason, the first step is to further define individual investment projects or project packages (measures) to be prioritized in the short term and developed further through project preparation for implementation. These are often referred to as 'bankable' projects[36]. For each of these projects it should be determined which is the most appropriate approach to follow. The best way is to see each project as a service that will be provided to

[36] A 'bankable' project is defined as a project or proposal that has sufficient collateral, future cashflow, and high probability of success, to be acceptable to institutional lenders for financing.

stakeholders. The main elements to be considered in the development of bankable projects and their required implementation arrangements are:

- The characteristics of the transaction: technical and financial characteristics of the project, such as how capital intensive the project is, how asset specific is the investment required (e.g. can the assets created be moved and reused for other purposes) and, most importantly, the main functions and services that will be provided through the investments as well as how these services can be classified in terms of economic good (private, common resource, club or public).
- The service level required over time.
- The institutional setting (stakeholders - strengths of local government, private sector and community – incentives created by formal and informal institutions).
- The necessary capacity development, as well as robust knowledge sharing between stakeholders and sectors, if applicable.

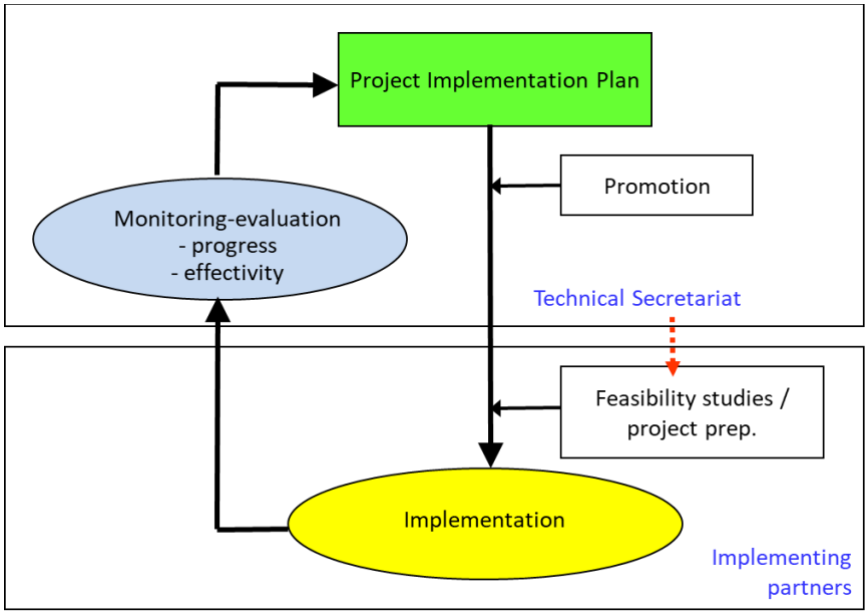
Based on these four elements the stakeholders in charge of the planning can choose from a wide range of project delivery and finance options that vary from solely public governance options up to the creation of markets for private initiatives by considering the options that would be most effective in ensuring sustainability in service provision. The ultimate result of this step will be a kind of Project Implementation Plan that describe the components of the project and the responsibilities of the implementing partners.

Implementation framework

While the actual implementation will be performed on a project level and may be undertaken by separate institutions (public and private), it remains necessary that the integrated character of the project from Phase III remains guarded. Individual components of the project might influence each other (positively and negatively), and the implementation needs to be monitored to take care that this mutual interference is tuned and that additional actions are taken if needed. For this an overarching implementation organization at national, provincial or river basin level might be necessary.

An overview of such overarching implementation organization is given in Figure 18. The actual implementation of most of the measures will take place by decentralized agencies of national ministries or at local governmental level and their related utilities, districts and associations. Where necessary, feasibility and engineering studies and Environmental and Social Impact Assessments (ESIA) should be carried out before the actual implementation and/or construction can take place.

Figure 18: Implementation Framework



The overarching implementation organization should include a Technical Secretariat (TS). Periodically, a monitoring report compiled by the TS can track the progress made in implementing the measures of the Project Implementation Plan and the effectiveness of these measures in meeting their objectives. Insufficient progress may lead to an adjustment of the Project Implementation Plan. The TS may also provide assistance to the implementing partners, e.g., the local government agencies, as they carry out feasibility studies. The TS should be able to support them by providing data and possibly other relevant information from their Management Information System (MIS).

3.6.2 Step IV.2 Funding and Financing of the Project and Procurement

With respect to funding and financing it is important that the difference between these terms is clear:

- Funding: how will the investments ultimately be paid for and/or have their costs recouped?
- Financing: how to make money available up-front for the investment? This is the money that has to be repaid over time (by the users or government).

Developing the financial arrangements for a project is a specialized activity which is beyond the scope of these water project design guidelines. Reference is made to handbooks on this subject such as the Handbook for the implementation of Nature-based Solutions for Water Security[37].

The **funding** of a project could be either public or private. In general terms the main sources of funding are what the OECD calls the 3 T's: Taxes, Tariffs or Transfers. The two main options to pay for the investments made are the taxpayers or the users. As water related infrastructure and water management projects are generally considered public goods, they are normally financed by public funds (taxes).

Depending on the type of project and whether the project sponsor is public or private, a variety of **financing** instruments could be used for mustering the up-front resources needed such as loans, bonds and others. These are to be repaid by the funding over time. Government departments can allocate budgets to the implementation. Beneficiaries might pay a tariff for the service introduced. Specific donors and/or international development agencies might be interested to fund the measure under specific models (e.g., 100% grant for part or full costs, or grant only for project preparation, etc.). The financial instruments of GCF are grants, loans, guarantees and equity.

[37] Altamirano, Monica A et al (2020). Handbook for the Implementation of Nature-based Solutions for Water Security: Guidelines for designing an implementation and financing arrangement

Funding and financing mechanisms can be both public and private. The blended finance approach provides the option to mix and blend all these different options of public and private flows of capital. “Blended finance”, defined by the OECD as “the strategic use of development finance and philanthropic funds to mobilize private capital flows to emerging and frontier markets”, can help in mobilizing private financing for water resource development.

As part of the financial analysis of a project, it is important to develop a budget plan for the project per component as well as an assessment of the social and economic costs and benefits of the project. When developing a financial analysis, it is important to analyse country specific indicators, such as GDP impact on project execution and success. A financial analysis is required for the entire project development including potential finance/role of the private sector and will be used by GCF to evaluate the level of concessionality on either a project level, per CO2, or per beneficiary. The project proposal should provide a strong economic and financial assessment of the project necessity to ensure a level of concessionality in line with the expected project performance when measured against GCF’s investment criteria.

Procurement refers to the way in which the project sponsor – a government agency or private actor – responsible for the project can choose to make use or purchase the different elements of the project. If the choice is to delegate to a third party, the project sponsor can tender it as a fully integrated contract – covering multiple life cycle phases- or choose to assign different phases and activities to different parties. Important considerations in procurement are the degree of risk transfer involved and incentives provided to the contracted party. Procurement processes can be lengthy and require additional capacity to be efficiently carried out. An astute approach might consider grouping different activities within a project in packages that can be procured efficiently.

It is noted that traditionally, the water sector is mostly funded by public institutions or governmental organizations. Due to climate finance, other financial players may enter the arena; local businesses for example also have a growing interest in safeguarding their own water supply under changing extreme conditions. A co-investment approach opens new opportunities for the business case for the implementation of measures.

Box 3. The case for Payment for Ecosystem Services in the Water Sector for implementing measures

Payment for ecosystem services differ from commercializing conventional commodities since it is not facilitated through a conventional commercial market. Therefore, the agreements and transactions are project specific. The beneficiaries of the project can be identified by analyzing regional value-chains, such as the water industry, for benefits coming from the regulation of ecosystem services. PES can be based on different mechanisms. For instance, since 2005 the English Water Company, Wessex Water, works with farmers in priority catchments to reduce pollutant loading in rivers. Removing pollutants at source is more cost-efficient to achieve compliance with drinking water standards than building complex water treatment facilities to process highly polluted water. Investments in natural flood management to complement ‘hard’ concrete flood walls are another example. For instance, England’s Countryside Stewardship programme provided financial incentives for land managers and farmers to deliver environmental benefits that included those in support of upstream measures that support flood risk management and conservation. The programme covered the costs for such changes.

3.6.3 Step IV.3 Prepare GCF funding proposal

The Guidelines presented in this document primarily address the technical content of the funding proposal package. Guidance for developing the funding proposal, as well as all other annexes required, are provided in the GCF Programming Manual[38]. The funding proposal must address the results of all phases as illustrated in Figure 4.

[38] https://www.greenclimate.fund/document/programming-manual

The structured approach will deliver dedicated output to the concept note (CN) and funding proposal (FP). Below a summary is provided which output can be used to inform which part of the CN and FP:

- Project/programme information (section B of FP, section B1-2 of CN) – This requires:
    - a description of the WRS (NRS, SES and AIS), policy context (institutional setting, general goals and policies, legal framework) provided by step I of the structured approach.
    - a description of the 'needs of the recipient' for adaptation and mitigation action. This is output provided by step II which also identifies in terms of the climate rationale which climate impacts are addressed by the project.
    - a description of the proposed project activities and how they respond to these climate impacts. This information results from step III.
- Financing information (section C of FP and CN) – This information is collected in step IV.2
- Expected performance against investment criteria (section D of FP and B3 of CN) Table 1 illustrates how almost all steps of the structured approach contribute to the different investment criteria.
- Theory of Change (section E of FP) - A primary activity that must be carried is to develop a Theory of Change for the proposed project. The Theory of Change should be developed by defining the project goals, output and outcomes, the necessary conditions to meet the project goals, and the barriers and risks to realize project outcomes. The Theory of Change should converge towards the GCF's governing instrument and the adaptation and/or mitigation logic for the project. This Theory of Change summarizes the results of the technical and content analysis of phases I to III and describes how the proposed project serves to shift the development pathway towards low-emission and/or climate-resilient development. The GCF Programming Manual (section B.2) describes the steps involved in developing the Theory of Change.

Before actual implementation of the formulated project and depending on the type and size of the project it might be necessary to carry out additional preparation activities such as a detailed feasibility study and engineering design work. For projects with significant environmental and social impacts, an Environmental and Social Impact Assessment (EISA) may be required. Environmental and social impacts may have already been considered in Phases I to III at a strategic level but, again depending on the project and the (country) context, a formal ESIA may be required. It is important to determine the Environmental and Social Safeguards categorisation of the proposed project as early as possible, as this will determine the applicable review process and public disclosure requirements.

3.7 Applying the structured approach for designing sub-sector projects

In designing projects for the sub-sectors IWRM, WASH and Drought and Flood Management the general approach of the four phases and steps as described in this document (Part 1) should be followed. Project conditions determine how much attention must be given to each of the steps and this will be different for each project. Specific recommendations are provided for the three sub-sectors in the separate Part 2 guidance document with respect to:

1. The impacts that Climate Change has on the sub-sector
2. The ambitions of GCF for the sub-sector
3. How to apply the structured approach for GCF funding projects
    3.1. The climate rationale
    3.2. The analytical tools (data and models) to be considered for the analysis
    3.3. The measures to be considered to increase the resilience

Criterion	Definition	Coverage area	Activity-specific sub-criteria	Indicative assessment factors (including indicators (both quantitative indicators and qualitative factors)
Impact potential	Potential of the programme/ project to contribute to the achievement of the Fund's objectives and result areas	Mitigation impact	Contribution to the shift to low-emission sustainable development pathways	* Expected tonnes of carbon dioxide equivalent (t CO2 eq) to be reduced or avoided (PMF-M Core 1) * Degree to which activity avoids lock-in of long-lived, high-emission infrastructure * Expected increase in the number of households with access to low-emission energy * Degree to which the programme/project supports the scaling up of low-emission energy in the affected region by addressing key barriers * Expected number of MW of low-emission energy capacity installed, generated and/or rehabilitated * Expected increase in the number of small, medium and large low-emission power suppliers (PMF-M 6.0 and related indicator(s)), and installed effective capacity * Expected decrease in energy intensity of buildings, cities, industries and appliances (PMF-M 7.0 and related indicator(s)) * Expected increase in the use of low-carbon transport (PMF-M 8.0 and related indicator(s)) * Expected improvement in the management of land or forest areas contributing to emission reductions (PMF-M 9.0 and related indicator(s)) * Expected improvement in waste management contributing to emission reductions (e.g. the change in the share of waste managed using low-carbon strategies and/or the change in the share of waste that is recovered through recycling and composting); and/or * Other relevant indicative assessment factors, taking into account the Fund's objectives, priorities and result areas, as appropriate on a case-by-case basis
		Adaptation impact	Contribution to increased climate-resilient sustainable development	* Expected total number of direct and indirect beneficiaries, (reduced vulnerability or increased resilience); number of beneficiaries relative to total population (PMF-A Core 1), particularly the most vulnerable groups * Degree to which the activity avoids lock-in of long-lived, climate-vulnerable infrastructure * Expected reduction in vulnerability by enhancing adaptive capacity and resilience for populations affected by the proposed activity, focusing particularly on the most vulnerable population groups and applying a gender-sensitive approach

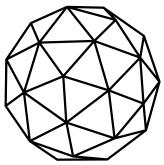
				* Expected strengthening of institutional and regulatory systems for climate-responsive planning and development (PMF-A 5.0 and related indicator(s)) * Expected increase in generation and use of climate information in decision-making (PMF-A 6.0 and related indicator(s)) * Expected strengthening of adaptive capacity and reduced exposure to climate risks (PMF-A 7.0 and related indicator(s)) * Expected strengthening of awareness of climate threats and risk-reduction processes (PMF-A 8.0 and related indicator(s)); and/or * Other relevant indicative assessment factors, taking into account the Fund's objectives, priorities and result areas, as appropriate on a case-by-case basis
Paradigm shift potential	Degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment	Potential for scaling up and replication, and its overall contribution to global low-carbon development pathways being consistent with a temperature increase of less than 2 degrees Celsius (mitigation only)	* Innovation * Level of contributions to global low-carbon development pathways, consistent with a temperature increase of less than 2 degrees Celsius * Potential for expanding the scale and impact of the proposed programme or project (scalability) * Potential for exporting key structural elements of the proposed programme or project elsewhere within the same sector as well as to other sectors, regions or countries (replicability)	* Opportunities for targeting innovative solutions, new market segments, developing or adopting new technologies, business models, modal shifts and/or processes * Expected contributions to global low-carbon development pathways consistent with a temperature increase of less than 2 degrees Celsius as demonstrated through: * A theory of change for scaling up the scope and impact of the intended project/programme without equally increasing the total costs of implementation * A theory of change for replication of the proposed activities in the project/programme in other sectors, institutions, geographical areas or regions, communities or countries
		Potential for knowledge and learning	Contribution to the creation or strengthening of knowledge, collective learning processes, or institutions	Existence of a monitoring and evaluation plan and a plan for sharing lessons learned so that they can be incorporated within other projects
		Contribution to the creation of an enabling environment	* Sustainability of outcomes and results beyond completion of the intervention * Market development and transformation	* Arrangements that provide for long-term and financially sustainable continuation of relevant outcomes and key relevant activities derived from the project/programme beyond the completion of the intervention * Extent to which the project/programme creates new markets and business activities at the local, national or international levels * Degree to which the activity will change incentives for market participants by reducing costs and risks, eliminating barriers to the deployment of low-carbon

				and climate-resilient solutions * Degree to which the proposed activities help to overcome systematic barriers to low-carbon development to catalyse impact beyond the scope of the project or programme
		Contribution to the regulatory framework and policies	Potential for strengthened regulatory frameworks and policies to drive investment in low-emission technologies and activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development	* Degree to which the project or programme advances the national/local regulatory or legal frameworks to systemically promote investment in low-emission or climate-resilient development * Degree to which the activity shifts incentives in favour of low-carbon and/or climate-resilient development or promotes mainstreaming of climate change considerations into policies and regulatory frameworks and decision-making processes at national, regional and local levels, including private-sector decision-making
		Overall contribution to climate-resilient development pathways consistent with a country's climate change adaptation strategies and plans (adaptation only)	* Potential for expanding the proposal's impact without equally increasing its cost base (scalability) * Potential for exporting key structural elements of the proposal to other sectors, regions or countries (replicability)	* Scaling up the scope and impact of the intended project/programme without equally increasing the total costs of implementation * A theory of change for replication of the proposed activities in the project/programme in other sectors, institutions, geographical areas or regions, communities or countries * Degree to which the programme or project reduces proposed risks of investment in technologies and strategies that promote climate resilience in developing countries
Sustainable development potential	Wider benefits and priorities	Environmental co-benefits	Expected positive environmental impacts, including in other result areas of the Fund, and/or in line with the priorities set at the national, local or sectoral level, as appropriate	Degree to which the project or programme promotes positive environmental externalities such as air quality, soil quality, conservation, biodiversity, etc.
		Social co-benefits	Expected positive social and health impacts, including in other result areas of the Fund, and/or in line with the priorities	Potential for externalities in the form of expected improvements, for women and men as relevant, in areas such as health and safety, access to education, improved regulation and/or cultural preservation

			set at the national, local or sectoral levels, as appropriate	
		Economic co-benefits	Expected positive economic impacts, including in other result areas of the Fund, and/or in line with the priorities set at the national, local or sectoral level, as appropriate	Potential for externalities in the form of expected improvements in areas such as expanded and enhanced job markets, job creation and poverty alleviation for women and men, increased and/or expanded involvement of local industries; increased collaboration between industry and academia; growth of private funds attracted; contribution to an increase in productivity and competitive capacity; improved sector income-generating capacity; contribution to an increase in energy security; change in water supply and agricultural productivity in targeted areas, etc.
		Gender-sensitive development impact	Potential for reduced gender inequalities in climate change impacts and/or equal participation by gender groups in contributing to expected outcomes	Explanation of how the project activities will address the needs of women and men in order to correct prevailing inequalities in climate change vulnerability and risks
Needs of the recipient	Vulnerability and financing needs of the beneficiary country and population	Vulnerability of the country (adaptation only)	Scale and intensity of exposure of people, and/or social or economic assets or capital, to risks derived from climate change	* Intensity of exposure to climate risks and the degree of vulnerability, including exposure to slow onset events * Size of population and/or social or economic assets or capital of the country exposed to climate change risks and impacts
		Vulnerable groups and gender aspects (adaptation only)	Comparably high vulnerability of the beneficiary groups	Proposed project/programme supports groups that are identified as particularly vulnerable in national climate or development strategies, with relevant sex disaggregation
		Economic and social development level of the country and the affected population	Level of social and economic development of the country and target population	Level of social and economic development (including income level) of the country and target population (e.g. minorities, disabled, elderly, children, female heads of households, indigenous peoples, etc.)
		Absence of alternative sources of financing	Opportunities for the Fund to overcome specific barriers to financing	Explanation of the existing barriers that create absence of alternative sources of financing and how they will be addressed
		Need for strengthening	Opportunities to strengthen institutional and	Potential of the proposed programme or project to strengthen institutional and implementation capacity

		institutions and implementation capacity	implementation capacity in relevant institutions in the context of the proposal	
Country ownership	Beneficiary country ownership of, and capacity to implement, a funded project or programme (policies, climate strategies and institutions)	Existence of a national climate strategy	*Objectives are in line with priorities in the country's national climate strategy * Proposed activity is designed in cognizance of other country policies	* Programme or project contributes to country's priorities for low-emission and climate-resilient development as identified in national climate strategies or plans, such as nationally appropriate mitigation actions (NAMAs), national adaptation plans (NAPs) or equivalent, and demonstrates alignment with technology needs assessments (TNAs), as appropriate * Degree to which the activity is supported by a country's enabling policy and institutional framework, or includes policy or institutional changes
		Coherence with existing policies		
		Capacity of accredited entities or executing entities to deliver	Experience and track record of the Accredited Entity or executing entities in key elements of the proposed activity	Proponent demonstrates a consistent track record and relevant experience and expertise in similar or relevant circumstances as described in the proposed project/programme (e.g., sector, type of intervention, technology, etc.)
		Engagement with civil society organizations and other relevant stakeholders	Stakeholder consultations and engagement	Proposal has been developed in consultation with civil society groups and other relevant stakeholders, with particular attention being paid to gender equality, and provides a specific mechanism for their future engagement in accordance with the Fund's environmental and social safeguards and stakeholder consultation guidelines. The proposal places decision-making responsibility with in-country institutions and uses domestic systems to ensure accountability
Efficiency and effectiveness	Economic and, if appropriate, financial soundness of the programme/project	Cost-effectiveness and efficiency regarding financial and non-financial aspects	* Financial adequacy and appropriateness of concessionality * Cost-effectiveness (mitigation only)	* Proposed financial structure (funding amount, financial instrument, tenor and term) is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers * Demonstration that the proposed financial structure provides the least concessionality needed to make the proposal viable * Demonstration that the Fund's support for the programme/project will not crowd out private and other public investment * Estimated cost per t CO2 eq (PMF-M Core 2) as defined as total investment cost/expected lifetime emission reductions, and relative to comparable opportunities

		Amount of co-financing	Potential to catalyse and/or leverage investment (mitigation only)	* Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (PMF-M Core 3) * Co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) * Potential to catalyse private- and public-sector investment, assessed in the context of performance on industry best practices * Expected indirect/long-term low-emission investment mobilized as a result of the implementation of activity
		Programme/project financial viability and other financial indicators	* Expected economic and financial internal rate of return * Financial viability in the long run	* Economic and financial rate of return with and without the Fund's support (i.e. hurdle rate of return or other appropriate/relevant thresholds) * Description of financial soundness in the long term (beyond the Fund's intervention)
		Industry best practices	Application of best practices and degree of innovation	* Explanations of how best available technologies and/or best practices, including those of indigenous peoples and local communities, are considered and applied * If applicable, the proposal specifies the innovations or modifications/adjustments made based on industry best practices



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